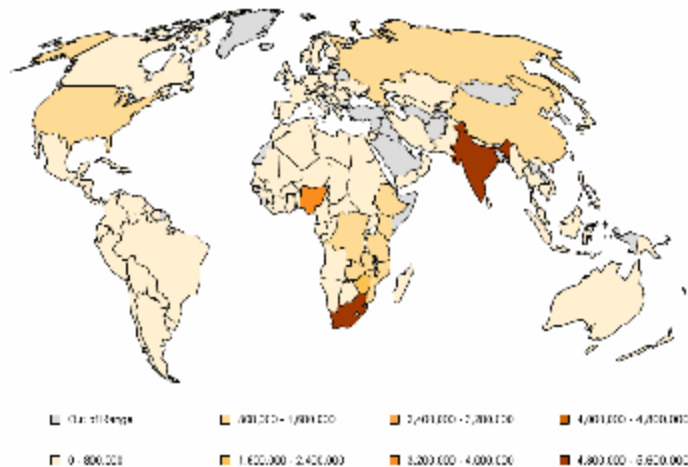


AIDS: A Global Crisis

Introduction

As the HIV/AIDS global enters its 25th years, both the number of infections and number of deaths due to the disease continue to rise. According to estimates from the UNAIDS/WHO AIDS Epidemic Update (December 2005), 38.0 million adults and 2.3 million children were living with HIV at the end of 2005. This is more than 50% higher than the figures projected by WHO in 1991 on the basis of the data then available.



(Source: <http://www.globalhealthfacts.org/topic.jsp?i=1#table>)

The overwhelming majority of people with HIV, some 95% of the global total, live in the developing world. The proportion is set to grow even further as infection rates continue to rise in countries where poverty, poor health care systems and limited resources for prevention and care fuel the spread of the virus.

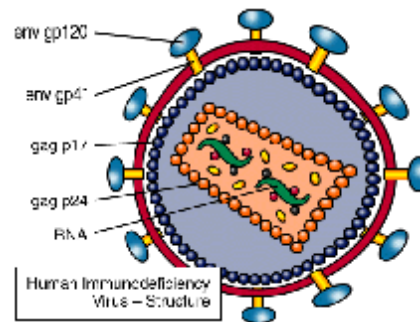
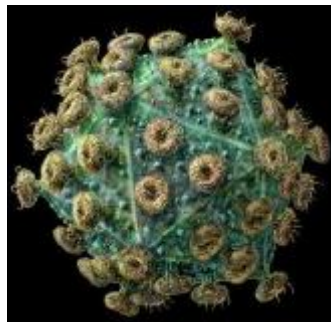
The area in Africa south of the Sahara desert, known as Sub-Saharan Africa, is by far the worst-affected in the world by the AIDS epidemic. The region has just over 10% of the world's population, but is home to over 60% of all people living with HIV.



(Source: http://www.unaids.org/bangkok2004/report_pdf.html)

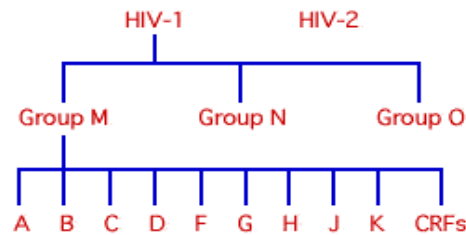
Around the world, the number of young people and children with HIV/AIDS and AIDS orphans has been rising up evidently. In Africa, there are amount of babies who acquire the virus at birth or through their mother's breast milk.

HIV stands for Human Immunodeficiency Virus. Like all viruses, HIV cannot grow or reproduce on its own. In order to make new copies of itself, it must infect the cells of a living organism.



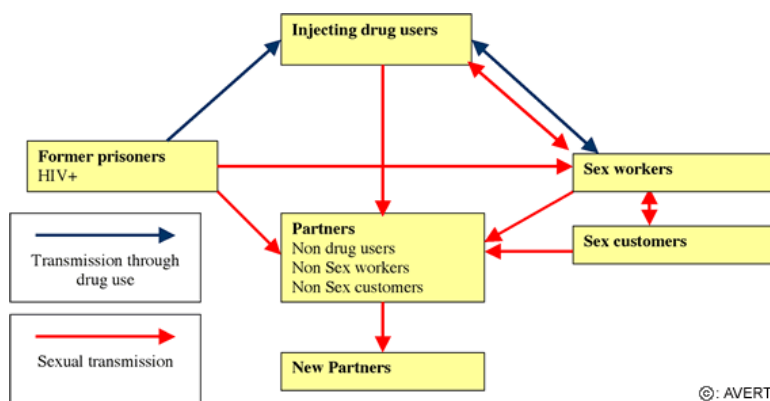
(Source: <http://www.avert.org/statindx.htm>)

Evidenced by the scientist, HIV can cause AIDS in about ten years. And HIV is a highly variable virus which mutates very readily. There are two types of HIV: HIV-1 and HIV-2.



(Source: <http://www.avert.org/statindx.htm>)

Both types are transmitted by injecting drug, sexual contact, through blood, and from mother to child, and they appear to cause clinically indistinguishable AIDS. However, it seems that HIV-2 is less easily transmitted, and the period between initial infection and illness is longer in the case of HIV-2.



(Source: <http://www.avert.org/statindx.htm>)

Future projections of the extent of the HIV/AIDS epidemic cannot be made with any precision; what happens next will depend on what action is taken. In some scenarios, governments and societies mount a very vigorous and wide-ranging response which recognises AIDS as much more than just a health issue, and so HIV prevalence eventually decreases; in other projections, good intentions fail to deliver anything more than short-term and fractured responses in the worst-affected countries, and the number of people living with HIV soars.

What are needed to turn the tide are massive responses at the national and international level:

- People need to challenge the myths and misconceptions about human sexuality that translate into dangerous sexual practices.

- Work and legislation is needed to reduce prejudice felt by HIV+ people around the world and the discrimination that prevents people from "coming out" as being HIV positive.
- HIV prevention initiatives need to be increased, people across the world need to be made aware of the dangers, the risks, and the ways they can protect themselves.
- Condom promotion and supply needs to be increased, and the appropriate sexual health education needs to be provided to young people before they reach an age where they become sexually active.
- Medication and support needs to be provided to people who are already HIV+, so that they can live longer and more productive lives, support their families, and avoid transmitting the virus onwards.
- Support and care needs to be provided for those children who have already been orphaned by AIDS, so that they can grow up safely, without experiencing poverty, exploitation, and themselves falling prey to HIV.

Assumptions and Hypothesis

1. In the model, when computing $\beta(t)$, we give the probabilities which means the possibility one suffer HIV through sex contact, injecting drug and so on by medical knowledge.

. Here, by medical knowledge, we assume the proportion in the table below

Table 1 proportion of each factor

Sex contact	0.08%	Injecting drug	50%
Perinatal	30%	Blood	50%

2. In the course of estimating, we suppose the number of people who are effective contacted by Sex contact, is 365 per year and others, we define 1.
3. The effective rate of ARV is 0.4.

Variable and Definition

t -- time

$S(t)$ -- number of susceptible individuals

$I(t)$ -- number of infected individuals without AIDS

$A(t)$ -- number of AIDS cases

$N(t)$ -- number of susceptible individuals

$u(t)$ -- death rate of individuals without AIDS ^[4]

$l(t)$ -- rate from susceptible individuals to infected individuals without AIDS

$d(t)$ -- death rate of AIDS

$\beta(t)$ -- effective contact rate

$g(t)$ -- rate from infected individuals without AIDS to AIDS

$a(t)$ -- birth rate in t year ^[birth_rate_data.xls]

Task I

Problem Analysis

The HIV that causes AIDS is primarily transmitted through sexual contact (man-woman, man-man), injecting drug use, Mother to infant, blood transfer and so on. According to these factors, we can estimate unknown data by the known facts, and for estimating it we need a mathematical model that considers relevant influencing factors.

First, we divide people into three ingredients: number of susceptible individuals, number of infected individuals without AIDS, and number of AIDS cases. Year by year, there are partial people of susceptible individuals transmitted infected individuals without AIDS, and a part of infected individuals without AIDS into AIDS cases.

Second, considering mother to infant, if one woman infected HIV, the baby conceived may be living with HIV. Hence there are some infants with the virus in the world.

In some developed countries, HIV spread in homosexual man, and the action turns out to be a large proportion, above all in recent years. And with development of level of economy, the phenomena may rise.

Many other factors can also affect the number of HIV or AIDS. But in the model we cannot consider any additional interventions and build a model to estimate the tendency.

Model design

Based on the analysis above, without considering any additional interventions, the AIDS expand freely. We present a simple model to predict the estimated rate of change in the number of HIV inflections. The model is given two forms by us.

Owing to existing transformation in different stages, such that susceptible individuals enter into infected individuals without AIDS, that is, the data of susceptible individuals at $t+1$ equals that the data of incoming of susceptible individuals at t minus injected data and the death; injected individuals without AIDS at $t+1$ equals injected individuals without AIDS at t adding the new incoming data minus the data of transfer to AIDS and the death; the number of AIDS at t equals the increased AIDS minus the death. Hence, we can establish the models following.

Form 1(SI model)

$$\begin{aligned}\frac{dI(t)}{dt} &= \beta(t)I(t)S(t) - lI \\ \frac{dA(t)}{dt} &= lI \\ S(t) + I(t) + A(t) &= P(t)\end{aligned}$$

Form 2

$$\begin{aligned}S(t+1) &= [1 - \alpha(t)][P(t+1) - P(t)] + [1 - \lambda(t) - u(t)]S(t) \\ I(t+1) &= \lambda(t)S(t) + [1 - \lambda(t) - u(t)]I(t) + \alpha(t)[P(t+1) - P(t)] \\ A(t+1) &= \gamma I(t) + [1 - \delta(t)]A(t) \\ S(t) + I(t) + A(t) &= P(t)\end{aligned}$$

In the above

$$\text{Infection rate} \quad \lambda(t) = \frac{\beta(t)I(t)}{S(t) + I(t)}$$

Rate of developing AIDS of infected individuals

$$g(t) = \frac{A(t+1) - A(t)}{I(t)}$$

Deaths rate $u(t)$ is different in distinct region, and at the present, there is no effective method to cure AIDS, and he will die in one year after examined with the

virus, so here we defined $d(t) = 1$.

In the course of estimating, if we face the proportion of man and woman with HIV is approximately equality, we can change the model above into:

$$\begin{aligned} S(t+1) &= P(t+1) - P(t) - I(t) * b(t) * \alpha(t) + [1 - \lambda(t) - u(t)]S(t) \\ I(t+1) &= \lambda(t)S(t) + [1 - \gamma(t) - u(t)]I(t) + I(t) * b(t) * \alpha(t) \\ A(t+1) &= \gamma I(t) + [1 - \delta(t)]A(t) \end{aligned}$$

The interrelated parameters are all the same.

Effective contact rate $\beta(t)$:

Here, we give a new method to establish $\beta(t)$. There are several factors to act on AIDS, and in different countries, the data is diverse. We define

n_i --ratio of the number of HIV by each factors and total number of HIV

k_i --the proportion that each action lead to HIV

Namely

$$\beta(t) = \sum_i n_i k_i$$

i is the amount of kinds about HIV, and is different in different countries. We take **India** for example.

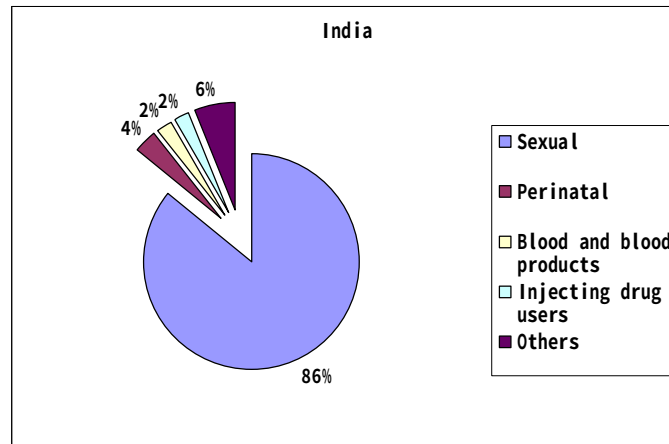


Figure 1 proportion of factors lead to HIV

According to the figure above, we can get the detailed proportion of the factors that result in HIV in India.

Using the model above and the assumptions ahead, we can work out the

Effective contact rate

$$\beta(t) = 86\% \times 0.08\% \times 365 + 4\% \times 30\% + 2\% \times 50\% + 2\% \times 50\% = 0.26$$

Then according to proportion of other countries, we can get a list form below

Table 2 proportion of HIV

Country	b	Country	b
US	0.26	Russia	0.5
India	0.26	Brazil	0.3464
Botswana	0.447	Australia	0.2

(the referent statistics come from <http://www.avert.org>)

Critical countries

We choose the critical country in each of the continents based on the known data and the structure of HIV in each country. Above all, according to effect from the country and the tendency, we choose the typical.

Africa—Botswana: Mother to infants has a sizable influence to the spread of HIV, and the total number of HIV takes up largely of the population.

Asia—India: Recent years, the number of HIV rises up quickly. From the known data, the number of HIV in India is 510 millions in 2003, and now it becomes the largest country that owns the number of AIDS except South Africa.

Europe-- Russian Federation: Injecting drugs and Sex deluge the spread of HIV in Russia, and more and more youths step into it. If not control strictly, the case will be serious in the future and it will threaten the government directly.

North America—US: With the development of economy, people are more exoteric about sex. So the Sex contact is the major ingredient that leads to HIV. According to statistics, among it, homoerotism man takes up largely than woman. If not take some measures, the number of HIV will rise up rapidly in the future.

Australia: Similarly to US, Sex contact is the major in the country, above all the homoerotism man.

South America—Brazil: It locates in a region with much drug around it, hence the amount of HIV rise up year by year because of injecting drug.

Results

Here, we define rate if HIV in each country per year to reflect the expected rate of change in the number of HIV/AIDS infections.

$$rate(k) = \frac{HIV(k)}{population(k)}$$

Where $HIV(k)$ is the number of HIV in k year; $population(k)$ is the total number of people of country in k year.

From the model we establish above, we can compute $HIV(k)$, and combined with $population(k)$, we can easily get the results.

Here we all estimate the data from 1999 year, and take the data in 1999 as initialization.

1. US

Initialization: $I(1999) = 850000$; $P(1999) = 281\text{millions}$; $\beta = 0.26$;

Replace the value in model with the initialization data, we can get the number of HIV/AIDS from 1999 to 2050 year and the rate of change in the following form.

Table 3 the number of HIV injection people and rate

Year	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
HIV	850000	978322	1125901	1295592	1490658	1714829	1972363	2268111	2607597	2997093
Rate(%)	0.30	0.34	0.39	0.45	0.51	0.58	0.66	0.75	0.86	0.98
Year	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
HIV	3443708	3955479	4541463	5211835	5977962	6852517	7849506	8984325	10273748	11735866
Rate(%)	1.11	1.27	1.44	1.64	1.87	2.12	2.41	2.74	3.10	3.52
Year	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
HIV	13389944	15256170	17355268	19707945	22334121	25251928	28476418	32017990	35880491	40059060
Rate(%)	3.99	4.51	5.09	5.74	6.46	7.26	8.13	9.09	10.12	11.23
Year	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038
HIV	44537768	49287204	54262216	59400125	64619741	69821586	74889713	79695433	84103057	87977465
Rate(%)	12.41	13.66	14.95	16.28	17.61	18.93	20.20	21.39	22.47	23.39
Year	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048
HIV	91192967	93642600	95246748	95959884	95774402	94720921	92865008	90300834	87142787	83516255
Rate(%)	24.14	24.67	24.99	25.07	24.91	24.54	23.96	23.21	22.31	21.30
Year	2049	2050								
HIV	79548786	75362544								
Rate(%)	20.22	19.08								

Based on the data expected, we plot the curve on the number of HIV/AIDS and the rate of change in the picture below

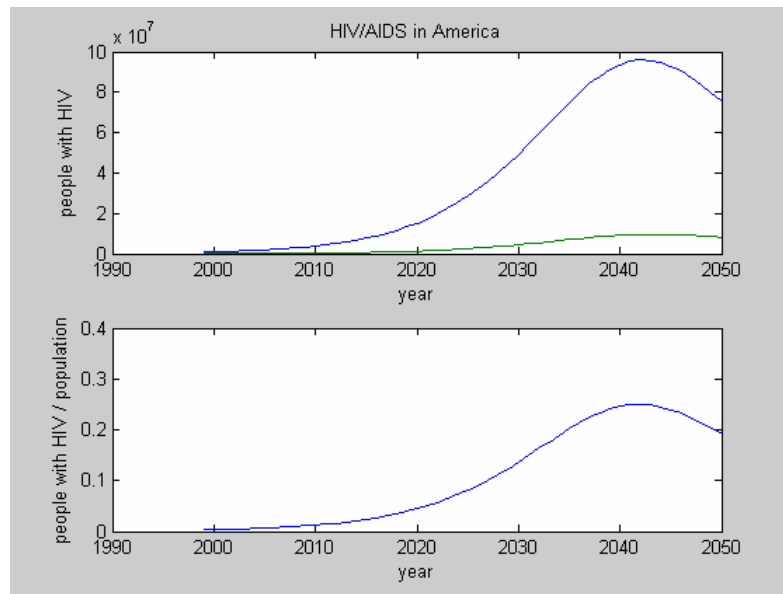


Figure 2 trend of the number of HIV in US

According to the figure, we can easily see the future trend, and the curve fits to common infection model. After suffering a peak, it will decrease gradually.

We can use $\frac{HIV(t+1) - HIV(t)}{HIV(t)} \times 100\%$ to compute the spread speed in HIV+ and it

can show the threaten of HIV/AIDS

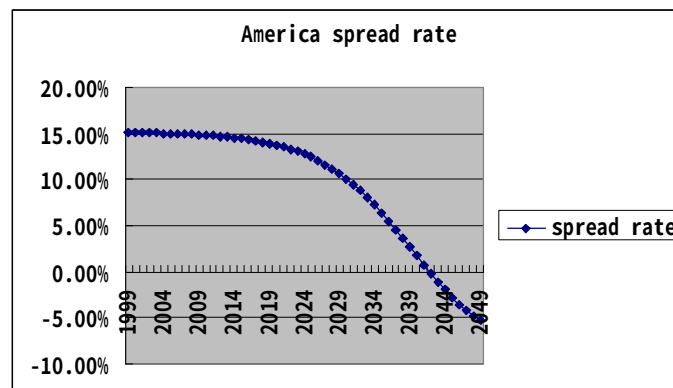


Figure 3 Rate of change in the number of HIV in US

2. India

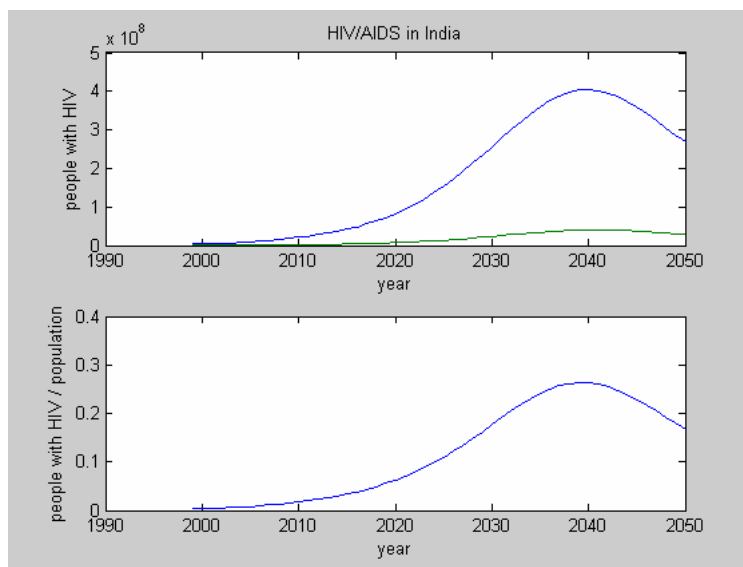
Initialization: $I(1999) = 3.7 \text{ millions}$; $P(1999) = 1001.4 \text{ millions}$; $\beta = 0.26$;

Replace the value in model with the initialization data, we can get the number of HIV from 1999 to 2050 year and the rate of change in the following form.

Table 4 the number of HIV injection people and rate

Year	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
HIV	3700000	4285035	4961965	5745022	6650574	7697422	8905813	11768143	13598893	15708599
Rate(%)	0.37	0.42	0.48	0.54	0.62	0.71	0.81	1.05	1.20	1.36
Year	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
HIV	18137838	20932450	24144011	27817160	32025370	36838872	42334591	48595912	55685382	63716272
Rate(%)	1.55	1.77	2.01	2.29	2.60	2.96	3.36	3.81	4.32	4.89
Year	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
HIV	72784265	82985299	94411942	107096227	1211478571	136613343	153508002	171805466	191337894	212034208
Rate(%)	5.52	6.23	7.02	7.88	8.83	9.87	11.00	12.21	13.49	14.84
Year	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038
HIV	233676023	255962838	278508480	300745527	322223148	342352476	360525828	376155004	388657358	397668363
Rate(%)	16.24	17.66	19.09	20.49	21.82	23.04	24.13	25.03	25.72	26.18
Year	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048
HIV	402902273	404231472	401695408	395424497	385845898	373409975	358634712	342064105	324160370	305505099
Rate(%)	26.39	26.35	26.06	25.53	24.81	23.91	22.88	21.74	20.53	19.29
Year	2049	2050								
HIV	286519820	267549753								
Rate(%)	18.04	16.80								

Similarly, based on the data expected, we plot the curve on the number of HIV/AIDS and the rate of change in the picture below

**Figure 4 trend of the number of HIV In India**

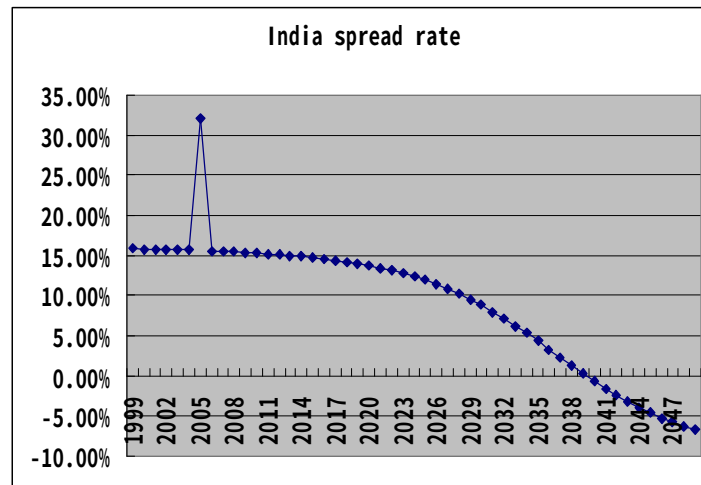


Figure 5 Rate of change in the number of HIV in US

From the figure, we can find the peak of HIV in India is about in 2040 year, and after that, according to the rule of infection model, the HIV will fall down.

In the same way, we also estimate HIV in another critical countries, but there are a lot of data and pictures, so we put them into appendix.

Task II

Problem Analysis

In **Task I**, from the estimated data, we can easily see that the number of HIV/AIDS will increase rapidly without any control. So we must take some measures.

Now, there are three scenarios: ARV drug therapy, a preventative HIV/AIDS vaccine and the both.

ARV drug therapy will keep back spread of HIV, and in the model it acts on effective contact rate. According to the statistics, the effective rate is 0.4, and it means that it can decrease 40% of the estimated HIV infection.

A preventative HIV/AIDS vaccine major fits to susceptible individuals. As the other vaccines, it can also keep people from HIV/AIDS and seldom infected if injecting it before infecting.

In this model, we ignore risk of emergence of drug-resistant of HIV and additional interventions.

Model design

Financial resources

In some regions, financial resources from foreign aid donors are very important, above all for some less developing countries. We find the historical data that the total number of financial resources in the world per year in the table below

Table 5 historical data of financial resources

Year	Total(\$)	Year	Total(\$)
1995	13096007	2000	67215179
1996	60192576	2001	74674761
1997	51011063	2002	91948663
1998	50685247	2003	122541801
1999	57003289	2004	149763125

(Source: <http://www.unaids.org/en/GetStarted/Donors.asp>)

According to the form above, we plot the picture that the total financial resources each year

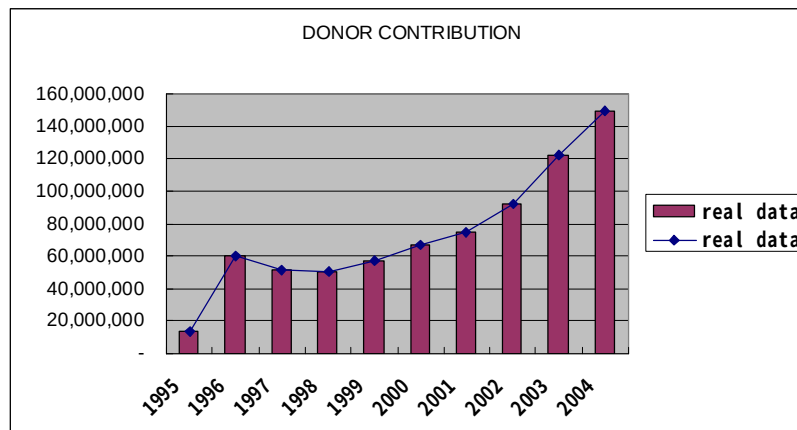


Figure 6 historical financial resources

From the figure, the total financials increase gradually year by year. This is mainly because HIV/AIDS crisis is strict and it is treated seriously in the world. According to the historical data, we also give the curve of the tend of total money by simulation .

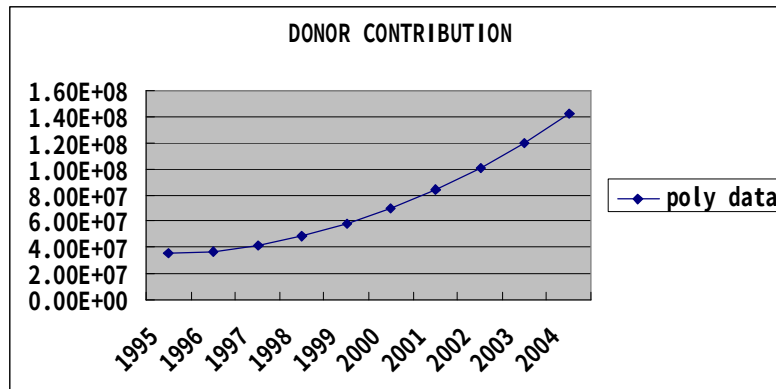


Figure 7 financial resources of estimated data by simulation

Then following the curve we poly above, we can give the value of total number from 2005 to 2050 by simulation.

Table 6 financial resources of estimated data

Year	Total(\$)	Year	Total(\$)	Year	Total(\$)	Year	Total(\$)
2005	166149567.07	2017	649769365.42	2029	1489799990.01	2041	2686241440.85
2006	192838303.43	2018	706159003.97	2030	1575890530.75	2042	2802032883.77
2007	222002114.97	2019	765023717.70	2031	1664456146.66	2043	2920299401.87
2008	253641001.70	2020	826363506.61	2032	1755496837.76	2044	3041040995.16
2009	287754963.61	2021	890178370.70	2033	1849012604.04	2045	3164257663.63
2010	324344000.70	2022	956468309.98	2034	1945003445.51	2046	3289949407.28
2011	363408112.97	2023	1025,233,324.44	2035	2043469362.15	2047	3418116226.11
2012	404947300.42	2024	1096473414.08	2036	2144410353.98	2048	3548758120.12
2013	448961563.05	2025	1170188578.90	2037	2247826420.99	2049	3681875089.32
2014	495450900.87	2026	1246378818.90	2038	2353717563.18	2050	3817467133.70
2015	544415313.87	2027	1325044134.09	2039	2462083780.55		
2016	595854802.06	2028	1406184524.46	2040	2572925073.11		

The actual number in 2005 is \$167544099, and the estimated number is \$166149567.03. They are basically equally.

Distribution of Financial resource

Considering world bank income code and the population of HIV/AIDS, we take 1999 year for example.

Table 7 the situation in 1999

The sum of donor contribution was \$57,003,289

Country	Income code	Reverse income code	HIV+
India	1	4	3700000
Brazil	3	1	540000
Australia	4	0	14000
America	4	0	850000
Russia	2	2	130000
Botswana	3	1	290000

Account all countries' reverse income codes

Use the following formula to compute the financial resource distribute to every country.

$$\frac{\text{ReverseIncomeCode} \times \text{HIV} +}{\sum \text{ReverseIncomeCode} \times \text{HIV} +} \times \text{FinancialResource}$$

We can account reverse income code use the following table

Reverse income code	Income code
4	1
2	2
1	3
0	4

According the situation of 1999, we can educe

Table 8 the distribution of financial resources

Country	Financial Resource
India	\$7,719,276.87
Brazil	\$281,649.29
Australia	0
America	0
Russia	\$135,608.92
Botswana	\$151,256.10

This distributing method can balance income level and population with HIV.

Model 1(ARV drug therapy)

According to the analysis above, we know effective rate is 0.4, and we define two parameters to reflect the relevancy:

conv - the rate of receiving ARVS in one country;

b - effective rate, and the number defines 0.4

And then

Effective contact rate changes into $\beta [1 - conv \times (1 - b)]$;

Rate from infected individuals without AIDS to AIDS changes into

$$\gamma [1 - conv \times (1 - b)]$$

Then use the model we established in the **Task I**, we can get the result. We also take Brazil for example.

Results

Brazil

We can get the relevant data in the known form^[2]. Here are

$$conv = 1; \quad b = 0.4$$

and other data about India is all the same.

Here, we consider intending to minimize the emergence of drug-resistant strains cost \$1100 per person per year.

We calculate the drug (annual) cost for patients as

$$\text{Drug: } (\text{the number of HIV}) \times (\$1100/\text{patient} - \text{year})$$

Then we can conclude the rate of change in the number of HIV/AIDS infections and the number of HIV/AIDS from 1999 to 2050, and how much is the country's payout can dominate the situation of HIV/AIDS.

Table 9 estimated data of Brazil with ARV

Year	HIV	rate	Costs
1999	540000	0.32%	\$0.00
2000	591852	0.34%	\$651,040,000.00
2001	648660	0.37%	\$713,530,000.00
2002	710895	0.40%	\$781,980,000.00
2003	779069	0.43%	\$856,980,000.00
2004	853742	0.46%	\$939,120,000.00
2005	935527	0.50%	\$1,029,100,000.00
2006	1024687	0.54%	\$1,127,200,000.00
2007	1122279	0.59%	\$1,234,500,000.00

2008	1229089	0.63%	\$1,352,000,000.00
2009	1345973	0.69%	\$1,480,600,000.00
2010	1473863	0.74%	\$1,621,200,000.00
2011	1613016	0.80%	\$1,774,300,000.00
2012	1765151	0.87%	\$1,941,700,000.00
2013	1931449	0.94%	\$2,124,600,000.00
2014	2113193	1.02%	\$2,324,500,000.00
2015	2311772	1.10%	\$2,542,900,000.00
2016	2527714	1.20%	\$2,780,500,000.00
2017	2763450	1.29%	\$3,039,800,000.00
2018	3020722	1.40%	\$3,322,800,000.00
2019	3301409	1.52%	\$3,631,600,000.00
2020	3607540	1.65%	\$3,968,300,000.00
2021	3940135	1.78%	\$4,334,100,000.00
2022	4302489	1.93%	\$4,732,700,000.00
2023	4697089	2.09%	\$5,166,800,000.00
2024	5126597	2.27%	\$5,639,300,000.00
2025	5593853	2.45%	\$6,153,200,000.00
2026	6100346	2.66%	\$6,710,400,000.00
2027	6650546	2.88%	\$7,315,600,000.00
2028	7247810	3.12%	\$7,972,600,000.00
2029	7895673	3.37%	\$8,685,200,000.00
2030	8597837	3.65%	\$9,457,600,000.00
2031	9355926	3.95%	\$10,292,000,000.00
2032	10175796	4.27%	\$11,193,000,000.00
2033	11061518	4.62%	\$12,168,000,000.00
2034	12017246	5.00%	\$13,219,000,000.00
2035	13047179	5.40%	\$14,352,000,000.00
2036	14153332	5.83%	\$15,569,000,000.00
2037	15341663	6.29%	\$16,876,000,000.00
2038	16616129	6.79%	\$18,278,000,000.00
2039	17980460	7.31%	\$19,779,000,000.00
2040	19438059	7.88%	\$21,382,000,000.00
2041	20989059	8.48%	\$23,088,000,000.00
2042	22638266	9.11%	\$24,902,000,000.00
2043	24387322	9.79%	\$26,826,000,000.00
2044	26236989	10.50%	\$28,861,000,000.00
2045	28186976	11.25%	\$31,006,000,000.00
2046	30231995	12.04%	\$33,255,000,000.00
2047	32372400	12.86%	\$35,610,000,000.00
2048	34603607	13.72%	\$38,064,000,000.00
2049	36919223	14.61%	\$40,611,000,000.00
2050	39310864	15.53%	\$43,242,000,000.00

For viewing the visual change, we plot the curve about the number of HIV/AIDS and rate of change in the number of HIV/AIDS infections below.

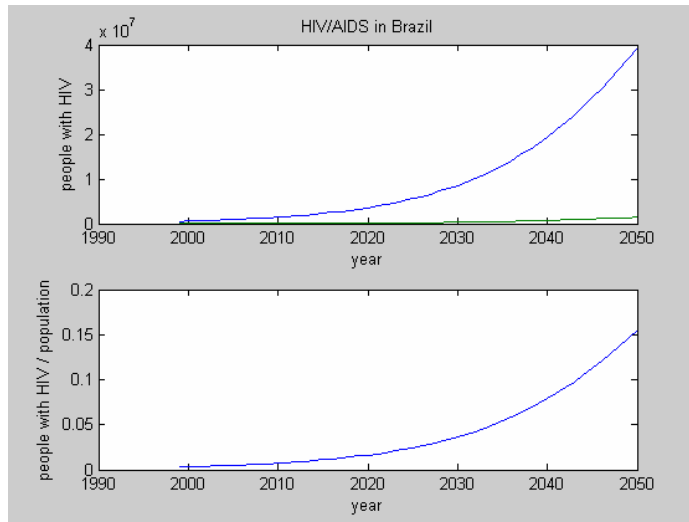


Figure 8 trend of the number of HIV with ARV In Brazil

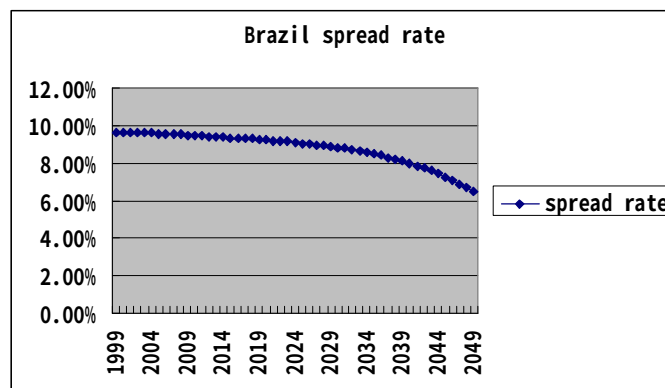


Figure 9 Rate of change in the number of HIV with ARV in Brazil

We also can draw the figure that before ARV control and compared with above figure.

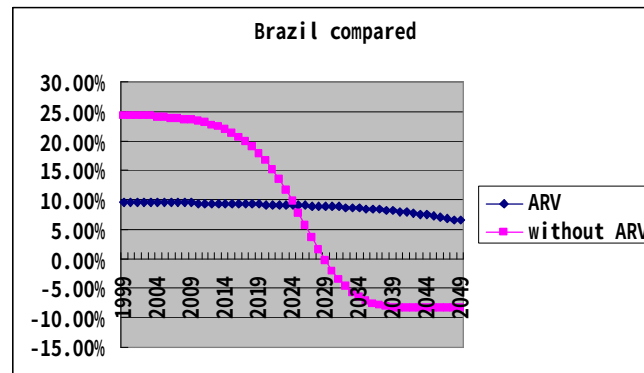


Figure 10 comparison of the two cases with ARV

And we can see that if not use ARV the spread rate stay a high level but will flow down in future. If use ARV it will make the spread rate keep a middle level, but will hold a long time to rise up the HIV+.

Model 2(HIV/AIDS vaccine)

Vaccine is aiming at susceptible individuals. So using vaccine can decrease the number of susceptible individuals. We also define a parameter to reflect the infection following.

s - vaccination coverage rate;

Noting: there are two vaccination coverage rate in the form, but we choose the higher data because it stands for the level which one country can present.

And then, there is a equation about susceptible individuals $S(t)$, where

$$S(t+1) = S(t)(1 - \sigma)$$

And combined the model we established in **Task I**, we can work out the number of HIV/AIDS and rate of change in the number of HIV/aids infections.

After vaccine appeared, it only cost \$0.75 per person, and it will reduce the cost largely. But according to scientists, the vaccine will appear after 2012 year at least. So we choose 2013 year as the dividing point which we will spend on vaccine in.

Total annual costs for vaccine are as follows:

$$\text{Vaccine: } (\text{the number of HIV}) \times (\$0.75/\text{patient} - \text{year})$$

Results

According to the analysis above, we introduce the model to work out the data following. We take Brazil for the example, here.

Table 10 estimated data of Brazil with vaccine

	HIV	rate	Costs
1999	540000	0.32%	\$0.00
2000	671330	0.39%	\$0.00
2001	834440	0.47%	\$0.00
2002	1036900	0.58%	\$0.00
2003	1288100	0.71%	\$0.00
2004	1599500	0.87%	\$0.00
2005	1985400	1.07%	\$0.00
2006	2461900	1.30%	\$0.00
2007	3050700	1.59%	\$0.00
2008	3776900	1.95%	\$0.00
2009	4671000	2.38%	\$0.00
2010	5769100	2.91%	\$0.00
2011	7110600	3.54%	\$0.00
2012	8746400	4.31%	\$127,410,000.00
2013	9206100	4.49%	\$5,657,000.00
2014	8363500	4.03%	\$1,648,500.00
2015	7533200	3.60%	\$1,514,300.00
2016	6782800	3.21%	\$1,485,500.00
2017	6109400	2.86%	\$1,455,500.00
2018	5505000	2.56%	\$1,425,900.00
2019	4962700	2.28%	\$1,397,000.00
2020	4475800	2.04%	\$1,368,500.00
2021	4037300	1.83%	\$1,340,900.00
2022	3643800	1.64%	\$1,311,500.00
2023	3290500	1.47%	\$1,281,300.00
2024	2973300	1.31%	\$1,250,000.00
2025	2688300	1.18%	\$1,217,600.00
2026	2431600	1.06%	\$1,185,700.00
2027	2200900	0.95%	\$1,152,600.00
2028	1993600	0.86%	\$1,117,000.00
2029	1807100	0.77%	\$1,078,200.00
2030	1639300	0.70%	\$1,037,200.00
2031	1487700	0.63%	\$995,130.00
2032	1351100	0.57%	\$953,450.00
2033	1227900	0.51%	\$914,100.00
2034	1116800	0.46%	\$878,290.00
2035	1016600	0.42%	\$845,080.00
2036	926020	0.38%	\$812,710.00
2037	844280	0.35%	\$779,370.00

2038	770430	0.31%	\$745,200.00
2039	703630	0.29%	\$709,640.00
2040	643130	0.26%	\$672,990.00
2041	588160	0.24%	\$635,950.00
2042	538240	0.22%	\$598,880.00
2043	492850	0.20%	\$562,150.00
2044	451520	0.18%	\$525,740.00
2045	413850	0.17%	\$489,380.00
2046	379390	0.15%	\$452,740.00
2047	347870	0.14%	\$415,330.00
2048	318970	0.13%	\$376,740.00
2049	292400	0.12%	\$336,550.00
2050	267900	0.11%	\$294,330.00

Plot the picture that describes the case following

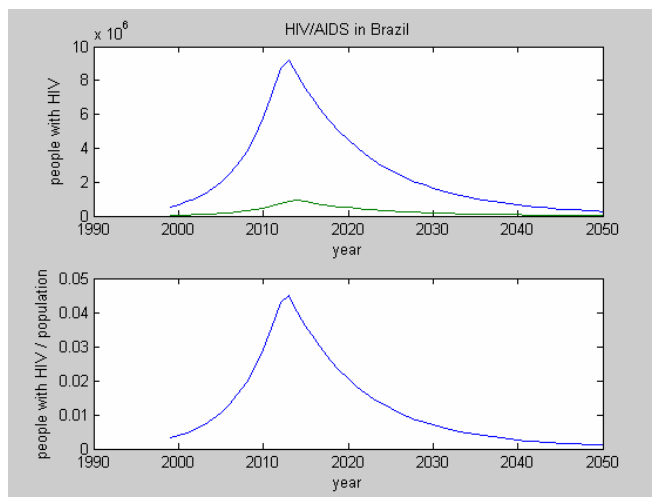


Figure 11 trend of the number of HIV In Brazil with vaccine

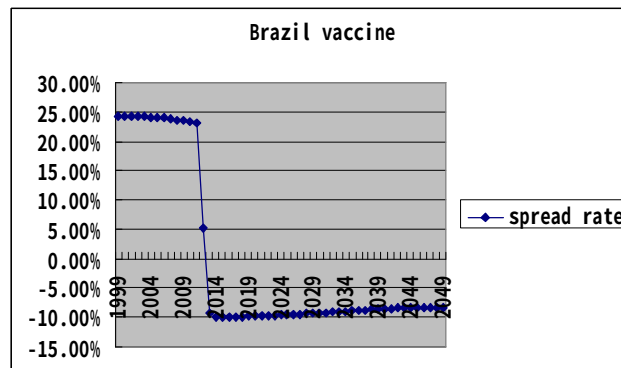


Figure 12 Rate of change in the number of HIV in Brazil with vaccine

From the figure, when vaccine come out after 2013 year, HIV/AIDS will be controlled clearly, and from it, the number of HIV/AIDS will decrease until a small part that easily dominated by the government.

And vaccine can reduce spread rate sharply (although it is joke that country can push vaccine over whole country after one year), we can consider that vaccine is far more effective than AVR.

Model 3(both)

Combining Model 1 and Model 2 above, we consider the influence of $conv$, b and δ , and the cost of both drug and vaccine. Using the basic model we established, we can also get the data.

Results

We take Brazil for example, and get

Table 11 estimated data of Brazil with both cases

	HIV	rate	Costs
1999	540000	0.32%	\$0.00
2000	591850	0.34%	\$651,040,000.00
2001	648660	0.37%	\$713,530,000.00
2002	710890	0.40%	\$781,980,000.00
2003	779070	0.43%	\$856,980,000.00
2004	853740	0.46%	\$939,120,000.00
2005	935530	0.50%	\$1,029,100,000.00
2006	1024700	0.54%	\$1,127,200,000.00
2007	1122300	0.59%	\$1,234,500,000.00
2008	1229100	0.63%	\$1,352,000,000.00
2009	1346000	0.69%	\$1,480,600,000.00
2010	1473900	0.74%	\$1,621,200,000.00
2011	1613000	0.80%	\$1,774,300,000.00
2012	1765200	0.87%	\$2,076,200,000.00
2013	1887300	0.92%	\$2,082,900,000.00
2014	1850000	0.89%	\$2,036,800,000.00
2015	1783400	0.85%	\$1,963,300,000.00
2016	1717300	0.81%	\$1,890,600,000.00
2017	1653900	0.77%	\$1,820,800,000.00
2018	1593000	0.74%	\$1,753,700,000.00
2019	1534500	0.71%	\$1,689,300,000.00
2020	1478200	0.67%	\$1,627,500,000.00
2021	1423800	0.64%	\$1,567,500,000.00

2022	1371400	0.62%	\$1,509,900,000.00
2023	1321200	0.59%	\$1,454,600,000.00
2024	1272900	0.56%	\$1,401,400,000.00
2025	1226400	0.54%	\$1,350,300,000.00
2026	1181500	0.51%	\$1,300,800,000.00
2027	1138200	0.49%	\$1,253,200,000.00
2028	1096600	0.47%	\$1,207,400,000.00
2029	1056600	0.45%	\$1,163,300,000.00
2030	1018000	0.43%	\$1,120,800,000.00
2031	980550	0.41%	\$1,079,600,000.00
2032	944460	0.40%	\$1,039,900,000.00
2033	909660	0.38%	\$1,001,600,000.00
2034	876110	0.36%	\$964,610,000.00
2035	843780	0.35%	\$929,020,000.00
2036	812500	0.33%	\$894,570,000.00
2037	782370	0.32%	\$861,390,000.00
2038	753330	0.31%	\$829,410,000.00
2039	725330	0.30%	\$798,580,000.00
2040	698320	0.28%	\$768,830,000.00
2041	672130	0.27%	\$739,990,000.00
2042	646850	0.26%	\$712,140,000.00
2043	622420	0.25%	\$685,230,000.00
2044	598820	0.24%	\$659,230,000.00
2045	576000	0.23%	\$634,100,000.00
2046	553860	0.22%	\$609,700,000.00
2047	532440	0.21%	\$586,110,000.00
2048	511710	0.20%	\$563,260,000.00
2049	491630	0.19%	\$541,130,000.00
2050	472140	0.19%	\$519,650,000.00

Plot the picture that describes the case following

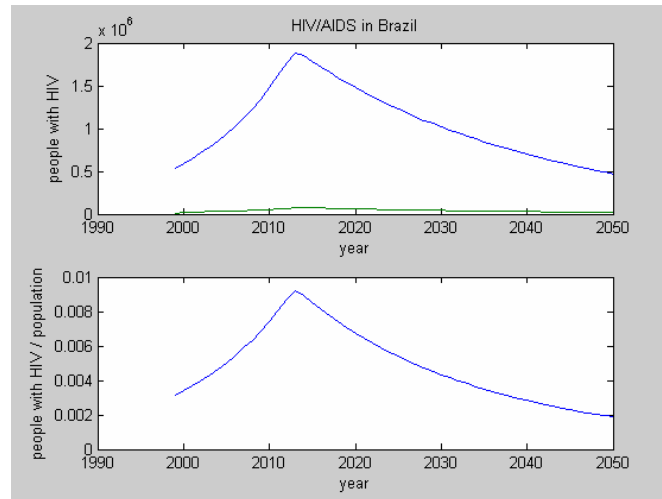


Figure 13 trend of the number of HIV In Brazil with both cases

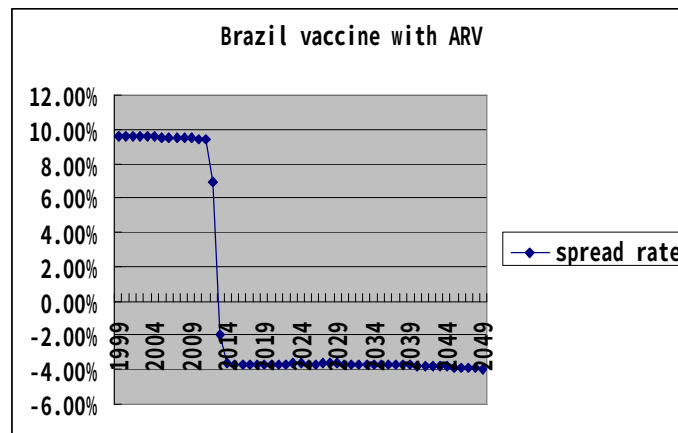


Figure 14 Rate of change in the number of HIV in Brazil with both

From the figure, we can conclude that when both act on HIV/AIDS, vaccine plays more.

Further discussion

Above, we take Brazil for example, and from the historical data, we can declare the rate of receiving ARVS in Brazil is very high, up to 100%. But in fact, the rate is much low in many other poor countries. In India, the number is only 2%. So for explaining better, we take India for example following and analysis the change if $conv$ is different.

First, when 2%, we compute the number of HIV/AIDS and the rate of change in the number of HIV/AIDS, and we give the curve below

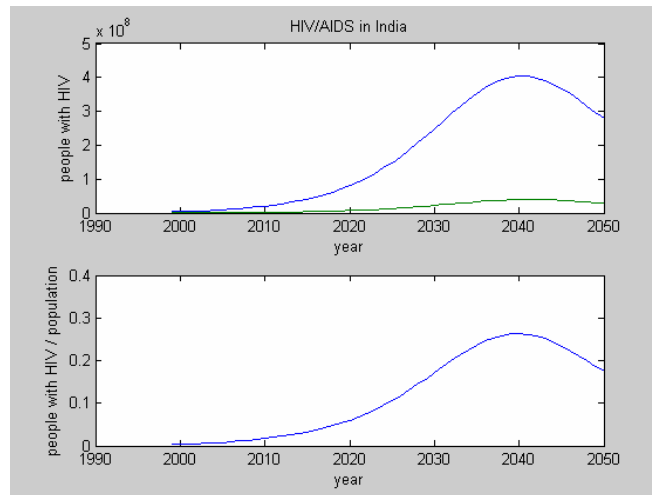


Figure 15 trend of the number of HIV in India when 2%

And then, we give the number from 10% to 100% and increases by 10%. We can see the change from the transformation. Similarly, we give the picture in the cases following

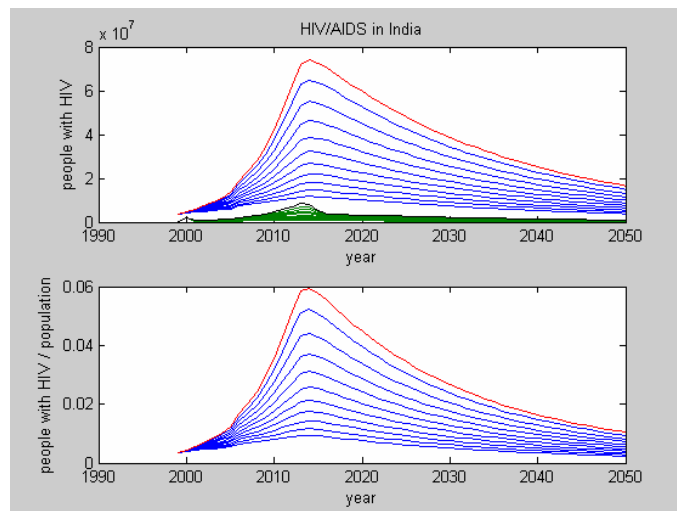


Figure 16 estimated trend when different rates

By comparing, from the two pictures above, we can easily conclude that with the increase of rate of receiving ARVS in India, the number of HIV/AIDS and rate of change of HIV/AIDS fall down. So ARVS is an effective measure to confine HIV before vaccine coming out. Then we should improve the coverage of ARVS in the world not only in development countries and it is good for human beings.

Task III

Problem Analysis

It always produces resistant if people take the same medicine frequently, and then, medicine will make no effect on virus. In the model, after taking medicine for a long time, we have to consider the factor.

Currently, when people infect AIDS, he can take two methods to prolong the life: ARV and vaccine (preventative). Major users are HIV infections. Here, we suppose that a person receiving ARV treatment with adherence below 90 percent has a 5 percent chance of producing a strain of HIV/AIDS which is resistance to standard first-line drug treatment.

Then we divide the infected individuals without AIDS $I(t)$ into three courses: no cure about it, being curing with ARV or vaccine and period with resistance to drug. So, according the model we established in **Task I**, we can get another model.

Model design

Based on the analysis above, considering the resistance to medicine, we present a model to predict the expected rate of change in the number of HIV infections.

According to **Task I**, we can establish a model below.

$$S(t+1) = P(t+1) - P(t) - [I_1(t) + I_2(t)] * b(t) * a(t) + [1 - l(t) - u(t)]S(t)$$

$$I_1(t+1) = l(t)S(t) + [1 - r_1 - u(t)]I_1(t) + I_1(t) * b(t) * a(t)$$

$$I_2(t+1) = r_1 I_1(t) + [1 - r_2 - u(t)]I_2(t) + I_2(t) * b(t) * a(t)$$

$$I_3(t+1) = r_2 I_2(t) + [1 - r - u(t)]I_3(t)$$

$$A(t+1) = r[I_1(t) + I_3(t)] + [1 - d(t)]A(t)$$

$$I_1(t) + I_2(t) + I_3(t) = I(t)$$

Where

$$r_1 = 0.9$$

$$r_2 = 0.05$$

$$r = 0.1$$

$$I_1(0) = I(0)$$

$$d(t) = 1$$

Here, we think over ARV-resistant and vaccine-resistant, and we take India for examples.

India

If virus has resistant to the medicine, this will affect the number of HIV/AIDS in the future. We estimate the number of people in the future and the rate of change in

HIV/AIDS.

1. ARV-resistant

Also in the data we use in **Task I**, we can get the number of HIV/AIDS from 1999 to 2050 year and the rate of change in the following form.

Table 12 estimated data of India under ARV-resistant

Year	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
HIV	3700000	4643500	5826600	7224900	8909500	10961000	13468000	18645000	2291000	28086000
Rate(%)	0.37	0.45	0.56	0.69	0.83	1.01	1.22	1.67	2.02	2.44
Year	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
HIV	34376000	42014000	51268000	62419000	75829000	91878000	11097000	133540000	159900000	190470000
Rate(%)	2.94	3.55	4.28	5.14	6.16	7.38	8.80	10.47	12.40	14.60
Year	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
HIV	225480000	265030000	308970000	356680000	407480000	460200000	513310000	565050000	613270000	656450000
Rate(%)	17.10	19.90	22.96	26.25	29.72	33.26	36.78	40.16	43.25	45.95
Year	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038
HIV	693110000	72221000	743200000	755830000	760800000	758800000	751050000	738370000	721790000	702390000
Rate(%)	48.16	49.84	50.95	51.49	51.51	51.08	50.26	49.14	47.77	46.24
Year	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048
AIDS	680990000	658280000	634810000	610900000	587010000	563390000	540200000	517580000	495500000	474120000
Rate(%)	44.60	42.90	41.18	39.45	37.74	36.08	34.46	32.89	31.38	29.93
Year	2049	2050								
HIV	453460000	43349000								
Rate(%)	28.55	27.22								

Based on the data expected, we plot the curve on the number of HIV/AIDS and the rate of change in the picture below

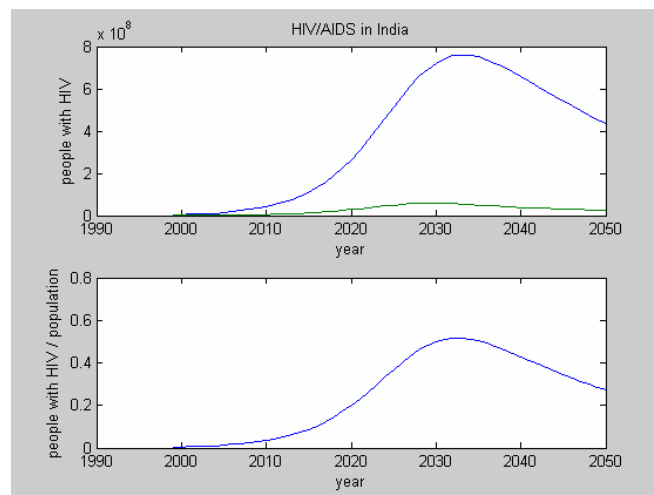


Figure 17 trend of the number of HIV in India under ARV-resistant

From the figure, we can conclude the amount AIDS now are more than AIDS expanding freely before.

2. Vaccine-resistant

Similarly, we can estimate the data of HIV from 1999 to 2050 in this case, and the data is in the form below

Table 13 estimated data of India under Vaccine-resistant

Year	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
HIV	3700000	4643500	5826600	7224900	8909500	10961000	13468000	18645000	22910000	28086000
Rate(%)	0.37%	0.45%	0.56%	0.69%	0.83%	1.01%	1.22%	1.67%	2.02%	2.44%
Year	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
HIV	34376000	42014000	51268000	62419000	71917000	73960000	72510000	70204000	67568000	64857000
Rate(%)	2.94%	3.55%	4.28%	5.14%	5.85%	5.94%	5.75%	5.51%	5.24%	4.97%
Year	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028
HIV	62175000	59571000	57066000	54638000	52317000	50101000	47984000	45961000	44006000	42137000
Rate(%)	4.72%	4.47%	4.24%	4.02%	3.82%	3.62%	3.44%	3.27%	3.10%	2.95%
Year	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038
HIV	40351000	38644000	37012000	35440000	33938000	32503000	31134000	29831000	28587000	27402000
Rate(%)	2.80%	2.67%	2.54%	2.41%	2.30%	2.19%	2.08%	1.99%	1.89%	1.80%
Year	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048
HIV	26273000	25195000	24164000	23173000	22222000	21310000	20436000	19596000	18785000	18003000
Rate(%)	1.72%	1.64%	1.57%	1.50%	1.43%	1.36%	1.30%	1.25%	1.19%	1.14%
Year	2049	2050								
HIV	17248000	16517000								
Rate(%)	1.09%	1.04%								

We plot a curve according to the data estimated,

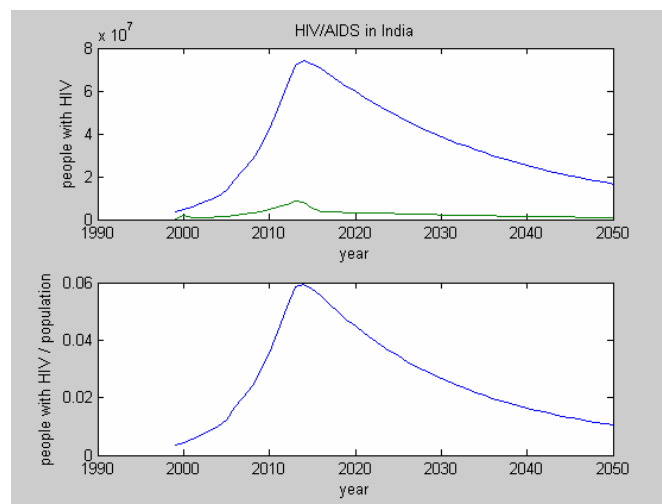


Figure 18 trend of the number of HIV in India under Vaccine-resistant

From the figure, we can conclude that Vaccine-resistant is less influence than

ARV-resistant.

Similarly, we can also conclude the HIV in other countries in the future, and these data will be found in appendix.

Depth model

In depth models, we construct a age groups models, with considering the difference between HIV1 and HIV2.

Variable and Definition

i -- the i age group

$S_i(t)$ -- number of susceptible individuals

$I_i(t)$ -- number of infected individuals without AIDS

$A_i(t)$ -- number of AIDS cases

$b_i(t)$ -- rate from susceptible individuals to infected individuals without AIDS

$g_i(t)$ -- rate from infected individuals without AIDS to AIDS

$m_i(t)$ --death rate

$F_i(t)$ --fertility of the women

$j_i(t)$ --the rate of woman in $S_i(t)$

$f_i(t)$ --the rate of woman in $I_i(t)$

$C(t)$ --the rate infected woman give birth a infected baby

$death_i(t)$ --the AIDS death rate

First we only consider the age groups' influence in the model.

We just base at the Task I 's model, and divide $S(t), I(t), A(t)$ into 101 age groups, for each age group we use the former model to analysis the question, and finally add them up to show the whole result.

$$\begin{aligned}
I_{i+1}(t+1) &= \beta_i(t)S_i(t) + I_i(t) - \gamma_i(t)I_i(t) - \mu_i(t)I_i(t); \\
S_{i+1}(t+1) &= (1 - \beta_i(t))S_i(t) - \mu_i(t)S_i(t); \\
S_\theta(t) &= \sum_{i=15}^{50} F_i(t)S_i(t)j_i(t) - \sum_{i=15}^{50} F_i(t)I_i(t)\varphi_i(t)C(t); \\
I_\theta(t) &= \sum_{i=15}^{50} F_i(t)I_i(t)\varphi_i(t)C(t); \\
A_{i+1}(t+1) &= A_i(t)(1 - death_i(t)) + I_i(t)\gamma_i(t);
\end{aligned}$$

For one year one group, divided a man's whole life into 101 year group 0..100;
 For a further considering, we add vaccine into upper model. (supposed that a man only can live 100 year for most)

$$\begin{aligned}
I_{i+1}(t+1) &= \beta_i(t)S_i(t) + I_i(t) - \gamma_i(t)I_i(t) - \mu_i(t)I_i(t); \\
S_{i+1}(t+1) &= (1 - \beta_i(t))S_i(t) - \mu_i(t)S_i(t); \\
S_\theta(t) &= \sum_{i=15}^{50} F_i(t)S_i(t)j_i(t) - \sum_{i=15}^{50} F_i(t)I_i(t)\varphi_i(t)C(t); \\
I_\theta(t) &= \sum_{i=15}^{50} F_i(t)I_i(t)\varphi_i(t)C(t); \\
A_{i+1}(t+1) &= A_i(t)(1 - death_i(t)) + I_i(t)\gamma_i(t); \\
S_{i+1}(t+1) &= S_{i+1}(t+1) - L_i(t) + L_{i-year}(t - year);
\end{aligned}$$

We add $L_i(t)$, it means the ability of vaccination in the country (the number of susceptible individuals that the country can vaccinate). And the vaccine has a period of validity named *year*, so we add the man who accept the vaccine but out of time $L_{i-year}(t - year)$ into $S_i(t)$.

Further more, the vaccines now on the way of researching are all for HIV1, so we considering the difference between HIV1 and HIV2.

The variable we add an upper mark HIV1 or HIV2 to different two types of HIV. In this model, the man with HIV1 also can infect HIV2, and the rate change to AIDS is high.

HIV1 is more epidemic than HIV2, it has higher infectivity, and the death rate is higher than HIV2. So HIV1 spread all over the world, but HIV2 is only epidemic in Africa, South America and so on.

So when the human develop the vaccine, we suppose the vaccine is for HIV1, for this reason, we divided the susceptible individuals into two types, accepted vaccine only prevent HIV1, and the whole susceptible individuals can infect HIV2.

$$I^{HIV1}_{i+1}(t+1) = b^{HIV1}_i(t)S^{HIV1}_i(t) + I^{HIV1}_i(t) - g^{HIV1}_i(t)I^{HIV1}_i(t) \\ - m_i(t)I^{HIV1}_i(t) - k^{HIV2}_i(t)I^{HIV1}_i(t);$$

$$I^{HIV2}_{i+1}(t+1) = b^{HIV2}_i(t)S_i(t) + I^{HIV2}_i(t) - g^{HIV2}_i(t)I^{HIV2}_i(t) \\ - m_i(t)I^{HIV2}_i(t) - k^{HIV1}_i(t)I^{HIV2}_i(t);$$

$$I^{HIV1\&2}_{i+1}(t+1) = k^{HIV1}_i(t)I^{HIV2}_i(t) + k^{HIV2}_i(t)I^{HIV1}_i(t) - g^{HIV1\&2}_i(t)I^{HIV1\&2}_i(t) \\ - m_i(t)I^{HIV1\&2}_i(t);$$

$$S_{i+1}(t+1) = (1 - b^{HIV2}_i(t))S_i(t) - m_i(t)S_i(t) - b^{HIV1}_i(t)S^{HIV1}_i(t);$$

$$S_0(t) = \sum_{i=15}^{50} F_i(t)S_i(t)j_i(t) - \sum_{i=15}^{50} F_i(t)I^{HIV1}_i(t)f_i(t)C^{HIV1}(t) \\ - \sum_{i=15}^{50} F_i(t)I^{HIV2}_i(t)f_i(t)C^{HIV2}(t) - \sum_{i=15}^{50} F_i(t)I^{HIV1\&2}_i(t)f_i(t)C^{HIV1\&2}(t);$$

$$I_0(t) = \sum_{i=15}^{50} F_i(t)I^{HIV1}_i(t)f_i(t)C^{HIV1}(t) + \sum_{i=15}^{50} F_i(t)I^{HIV2}_i(t)f_i(t)C^{HIV2}(t) \\ + \sum_{i=15}^{50} F_i(t)I^{HIV1\&2}_i(t)f_i(t)C^{HIV1\&2}(t)$$

$$A_{i+1}(t+1) = A_i(t)(1 - death_i(t)) + I^{HIV1}_i(t)g^{HIV1}_i(t) \\ + I^{HIV2}_i(t)g^{HIV2}_i(t) + I^{HIV1\&2}_i(t)g^{HIV1\&2}_i(t);$$

$$S^{HIV1}_{i+1}(t+1) = S^{HIV1}_{i+1}(t+1) - L_i(t) + L_{i-year}(t - year)$$

the first year to use HIV1 vaccine

$$S^{HIV1}_i(t) = S_i(t) - L_i(t) - b^{HIV2}_i(t)S_i(t), i = 0..100;$$

Additional, we also can consider this model with accepting AVR, HAART, and drug resistant of HIV. Use age group model can forecast the spread of HIV more exactly, but it needs more data for compute.

Strengths

1. We have discussed various influential factors such as the rate of receiving ARVS and vaccination coverage rate. So our model can be used widely without fundamental changes. Furthermore, some of our estimated data adapt to the actual data.
2. We have discussed the case of different rates of receiving ARVS and forecast the future. And so is availability for government to make decisions.
3. Above all, we put forward some detailed advices for controlling HIV/AIDS, and from the historical and estimated data we established, the advices do play a important role.
4. Especially, we have set up models ourselves, which is successful in discussing effective contact rate, financial resource and rate of number of HIV.

Weaknesses

We cannot find much data on the parameters needed, such as Effective contact rate. So we have to determine just a few values referring to some published data, which may be not representative. But it can be easily solved given practical data.

Reference

- [1] <http://www.avert.org/>
- [2] <http://www.who.int/GlobalAtlas.com>
- [3] <http://www.unaids.org/en/GetStarted/Donors.asp>

Appendix 1

Noting: rate is the number of HIV to the number of population ratio.

1.Uncontrolled Situation:

	India		Brazil		Australia	
	population	rate	population	rate	population	rate
1999	3700000	0.37%	540000	0.32%	14000	0.07%
2000	4285035	0.42%	671333	0.39%	15294	0.08%
2001	4961965	0.48%	834437	0.47%	16707	0.09%
2002	5745022	0.54%	1036907	0.58%	18250	0.09%
2003	6650574	0.62%	1288105	0.71%	19936	0.10%
2004	7697422	0.71%	1599546	0.87%	21777	0.11%
2005	8905813	0.81%	1985350	1.07%	23788	0.12%
2006	11768143	1.05%	2461914	1.30%	25984	0.13%
2007	13598893	1.20%	3050689	1.59%	28382	0.14%
2008	15708599	1.36%	3776943	1.95%	31001	0.15%
2009	18137838	1.55%	4671024	2.38%	33860	0.16%
2010	20932450	1.77%	5769055	2.91%	36983	0.17%
2011	24144011	2.01%	7110566	3.54%	40392	0.19%
2012	27817160	2.29%	8746436	4.31%	44114	0.20%
2013	32025370	2.60%	10732210	5.23%	48178	0.22%
2014	36838872	2.96%	13129260	6.33%	52613	0.24%
2015	42334591	3.36%	16002879	7.64%	57455	0.26%
2016	48595912	3.81%	19411925	9.18%	62740	0.28%
2017	55685382	4.32%	23420812	10.97%	68508	0.30%
2018	63716272	4.89%	28075291	13.03%	74802	0.33%
2019	72784265	5.52%	33395492	15.37%	81670	0.35%
2020	82985299	6.23%	39361419	17.96%	89164	0.38%
2021	94411942	7.02%	45884142	20.76%	97338	0.41%
2022	107096227	7.88%	52825847	23.71%	106254	0.45%
2023	121147857	8.83%	59958691	26.70%	115977	0.48%
2024	136613343	9.87%	66976240	29.60%	126580	0.52%
2025	153508002	11.00%	73511877	32.25%	138139	0.57%
2026	171805466	12.21%	79156133	34.48%	150738	0.61%
2027	191337894	13.49%	83571253	36.16%	164468	0.67%
2028	212034208	14.84%	86488158	37.18%	179426	0.72%
2029	233676023	16.24%	87770690	37.49%	195719	0.78%
2030	255962838	17.66%	87434583	37.13%	213461	0.85%
2031	278508480	19.09%	85611943	36.15%	232773	0.92%
2032	300745527	20.49%	82588017	34.68%	253789	0.99%
2033	322223148	21.82%	78666537	32.86%	276650	1.08%
2034	342352476	23.04%	74152380	30.82%	301507	1.17%
2035	360525828	24.13%	69317765	28.68%	328523	1.26%

2036	376155004	25.03%	64372235	26.51%	357871	1.37%
2037	388657358	25.72%	59497693	24.40%	389735	1.48%
2038	397668363	26.18%	54805560	22.38%	424311	1.60%
2039	402902273	26.39%	50365069	20.49%	461804	1.74%
2040	404231472	26.35%	46213202	18.73%	502434	1.88%
2041	401695408	26.06%	42357317	17.11%	546426	2.04%
2042	395424497	25.53%	38806297	15.62%	594022	2.20%
2043	385845898	24.81%	35548261	14.27%	645467	2.38%
2044	373409975	23.91%	32566146	13.03%	701017	2.58%
2045	358634712	22.88%	29840217	11.91%	760934	2.78%
2046	342064105	21.74%	27345655	10.89%	825482	3.01%
2047	324160370	20.53%	25067444	9.96%	894930	3.25%
2048	305505099	19.29%	22984385	9.11%	969540	3.50%
2049	286519820	18.04%	21076310	8.34%	1049570	3.77%
2050	267549753	16.80%	19324189	7.63%	1135268	4.06%

	America		Russia		Botswana	
	population	rate	population	rate	population	rate
1999	850000	0.30%	130000	0.09%	290000	16.71%
2000	978322	0.34%	180450	0.12%	365900	20.86%
2001	1125901	0.39%	250450	0.17%	454505	25.74%
2002	1295592	0.45%	347530	0.24%	553363	31.25%
2003	1490658	0.51%	482120	0.33%	656960	37.08%
2004	1714829	0.58%	668570	0.46%	756321	42.75%
2005	1972363	0.66%	926660	0.65%	839897	47.59%
2006	2268111	0.75%	1283700	0.90%	895706	50.90%
2007	2607597	0.86%	1776400	1.25%	916444	52.28%
2008	2997093	0.98%	2454600	1.74%	900964	51.61%
2009	3443708	1.11%	3384500	2.41%	854972	49.20%
2010	3955479	1.27%	4652900	3.32%	788275	45.58%
2011	4541463	1.44%	6369900	4.57%	710899	41.31%
2012	5211832	1.64%	8669800	6.25%	631120	36.86%
2013	5977962	1.87%	11705000	8.48%	554206	32.52%
2014	6852517	2.12%	15626000	11.37%	486555	28.67%
2015	7849506	2.41%	20544000	15.03%	427162	25.27%
2016	8984325	2.74%	26453000	19.45%	374945	22.25%
2017	10273748	3.10%	33147000	24.50%	329110	19.58%
2018	11735866	3.52%	40115000	29.81%	288879	17.22%
2019	13389944	3.99%	46523000	34.76%	253566	15.14%
2020	15256170	4.51%	51365000	38.59%	222569	13.32%
2021	17355268	5.09%	53802000	40.65%	195291	11.71%
2022	19707945	5.74%	53568000	40.71%	171356	10.29%
2023	22334121	6.46%	51049000	39.03%	150355	9.05%
2024	25251928	7.26%	47033000	36.17%	131927	7.96%

2025	28476418	8.13%	42315000	32.74%	115758	6.99%
2026	32017990	9.09%	37589000	29.27%	101530	6.15%
2027	35880491	10.12%	33390000	26.16%	89051	5.40%
2028	40059060	11.23%	29660000	23.38%	78106	4.74%
2029	44537768	12.41%	26347000	20.90%	68506	4.17%
2030	49287204	13.66%	23403000	18.67%	60086	3.66%
2031	54262216	14.95%	20793000	16.69%	52686	3.21%
2032	59400125	16.28%	18474000	14.92%	46198	2.82%
2033	64619741	17.61%	16413000	13.33%	40508	2.48%
2034	69821586	18.93%	14583000	11.92%	35520	2.17%
2035	74889713	20.20%	12956000	10.65%	31145	1.91%
2036	79695433	21.39%	11514000	9.52%	27305	1.67%
2037	84103057	22.47%	10232000	8.50%	23938	1.47%
2038	87977465	23.39%	9093100	7.60%	20987	1.29%
2039	91192967	24.14%	8080800	6.79%	18399	1.13%
2040	93642600	24.67%	7181300	6.07%	16131	0.99%
2041	95246748	24.99%	6382300	5.42%	14254	0.87%
2042	95959884	25.07%	5672200	4.85%	12831	0.79%
2043	95774402	24.91%	5041200	4.33%	11846	0.72%
2044	94720921	24.54%	4480300	3.87%	11255	0.69%
2045	92865008	23.96%	3981900	3.46%	11005	0.67%
2046	90300834	23.21%	3538400	3.09%	11048	0.67%
2047	87142787	22.31%	3144300	2.76%	11351	0.69%
2048	83516255	21.30%	2794200	2.47%	11888	0.72%
2049	79548786	20.22%	2483000	2.21%	12643	0.76%
2050	75362544	19.08%	2206500	1.97%	13606	0.82%

2. Controlled Situation:**Brazil 100%**

	pop	rate	money
1999	540000	0.32%	\$0.00
2000	591852	0.34%	\$651,040,000.00
2001	648660	0.37%	\$713,530,000.00
2002	710895	0.40%	\$781,980,000.00
2003	779069	0.43%	\$856,980,000.00
2004	853742	0.46%	\$939,120,000.00
2005	935527	0.50%	\$1,029,100,000.00
2006	1024687	0.54%	\$1,127,200,000.00
2007	1122279	0.59%	\$1,234,500,000.00
2008	1229089	0.63%	\$1,352,000,000.00
2009	1345973	0.69%	\$1,480,600,000.00
2010	1473863	0.74%	\$1,621,200,000.00
2011	1613016	0.80%	\$1,774,300,000.00
2012	1765151	0.87%	\$1,941,700,000.00

2013	1931449	0.94%	\$2,124,600,000.00
2014	2113193	1.02%	\$2,324,500,000.00
2015	2311772	1.10%	\$2,542,900,000.00
2016	2527714	1.20%	\$2,780,500,000.00
2017	2763450	1.29%	\$3,039,800,000.00
2018	3020722	1.40%	\$3,322,800,000.00
2019	3301409	1.52%	\$3,631,600,000.00
2020	3607540	1.65%	\$3,968,300,000.00
2021	3940135	1.78%	\$4,334,100,000.00
2022	4302489	1.93%	\$4,732,700,000.00
2023	4697089	2.09%	\$5,166,800,000.00
2024	5126597	2.27%	\$5,639,300,000.00
2025	5593853	2.45%	\$6,153,200,000.00
2026	6100346	2.66%	\$6,710,400,000.00
2027	6650546	2.88%	\$7,315,600,000.00
2028	7247810	3.12%	\$7,972,600,000.00
2029	7895673	3.37%	\$8,685,200,000.00
2030	8597837	3.65%	\$9,457,600,000.00
2031	9355926	3.95%	\$10,292,000,000.00
2032	10175796	4.27%	\$11,193,000,000.00
2033	11061518	4.62%	\$12,168,000,000.00
2034	12017246	5.00%	\$13,219,000,000.00
2035	13047179	5.40%	\$14,352,000,000.00
2036	14153332	5.83%	\$15,569,000,000.00
2037	15341663	6.29%	\$16,876,000,000.00
2038	16616129	6.79%	\$18,278,000,000.00
2039	17980460	7.31%	\$19,779,000,000.00
2040	19438059	7.88%	\$21,382,000,000.00
2041	20989059	8.48%	\$23,088,000,000.00
2042	22638266	9.11%	\$24,902,000,000.00
2043	24387322	9.79%	\$26,826,000,000.00
2044	26236989	10.50%	\$28,861,000,000.00
2045	28186976	11.25%	\$31,006,000,000.00
2046	30231995	12.04%	\$33,255,000,000.00
2047	32372400	12.86%	\$35,610,000,000.00
2048	34603607	13.72%	\$38,064,000,000.00
2049	36919223	14.61%	\$40,611,000,000.00
2050	39310864	15.53%	\$43,242,000,000.00

Russia 83.3%

	pop	rate	money
1999	130000	0.09%	\$0.00
2000	154490	0.11%	\$141,560,000.00
2001	183580	0.13%	\$168,220,000.00

2002	218150	0.15%	\$199,890,000.00
2003	259210	0.18%	\$237,510,000.00
2004	307970	0.21%	\$282,190,000.00
2005	365870	0.26%	\$335,250,000.00
2006	434740	0.31%	\$398,350,000.00
2007	516490	0.36%	\$473,260,000.00
2008	613530	0.43%	\$562,180,000.00
2009	728670	0.52%	\$667,680,000.00
2010	865230	0.62%	\$792,810,000.00
2011	1027100	0.74%	\$941,090,000.00
2012	1218800	0.88%	\$1,116,700,000.00
2013	1445700	1.05%	\$1,324,700,000.00
2014	1714000	1.25%	\$1,570,500,000.00
2015	2031000	1.49%	\$1,861,000,000.00
2016	2404400	1.77%	\$2,203,200,000.00
2017	2844200	2.10%	\$2,606,100,000.00
2018	3360900	2.50%	\$3,079,600,000.00
2019	3966600	2.96%	\$3,634,600,000.00
2020	4674500	3.51%	\$4,283,300,000.00
2021	5497800	4.15%	\$5,037,700,000.00
2022	6452200	4.90%	\$5,912,200,000.00
2023	7552800	5.77%	\$6,920,600,000.00
2024	8813700	6.78%	\$8,076,000,000.00
2025	10247000	7.93%	\$9,389,800,000.00
2026	11862000	9.24%	\$10,869,000,000.00
2027	13661000	10.70%	\$12,517,000,000.00
2028	15637000	12.33%	\$14,328,000,000.00
2029	17774000	14.10%	\$16,287,000,000.00
2030	20041000	15.99%	\$18,363,000,000.00
2031	22392000	17.98%	\$20,518,000,000.00
2032	24761000	20.00%	\$22,689,000,000.00
2033	27067000	21.99%	\$24,801,000,000.00
2034	29218000	23.88%	\$26,773,000,000.00
2035	31120000	25.58%	\$28,515,000,000.00
2036	32688000	27.02%	\$29,952,000,000.00
2037	33845000	28.13%	\$31,012,000,000.00
2038	34541000	28.87%	\$31,650,000,000.00
2039	34759000	29.21%	\$31,850,000,000.00
2040	34514000	29.17%	\$31,625,000,000.00
2041	33848000	28.76%	\$31,015,000,000.00
2042	32820000	28.04%	\$30,073,000,000.00
2043	31499000	27.06%	\$28,863,000,000.00
2044	29960000	25.88%	\$27,452,000,000.00
2045	28268000	24.56%	\$25,902,000,000.00

2046	26533000	23.18%	\$24,312,000,000.00
2047	24904000	21.89%	\$22,820,000,000.00
2048	23375000	20.66%	\$21,419,000,000.00
2049	21940000	19.51%	\$20,104,000,000.00
2050	20593000	18.43%	\$18,870,000,000.00

Australia 53.2%

	pop	rate	money
1999	14000	0.07%	\$0.00
2000	14848	0.08%	\$8,688,800.00
2001	15746	0.08%	\$9,214,700.00
2002	16699	0.09%	\$9,772,500.00
2003	17710	0.09%	\$10,364,000.00
2004	18782	0.09%	\$10,991,000.00
2005	19918	0.10%	\$11,656,000.00
2006	21123	0.10%	\$12,361,000.00
2007	22401	0.11%	\$13,109,000.00
2008	23756	0.11%	\$13,902,000.00
2009	25193	0.12%	\$14,743,000.00
2010	26716	0.13%	\$15,634,000.00
2011	28331	0.13%	\$16,579,000.00
2012	30043	0.14%	\$17,581,000.00
2013	31859	0.15%	\$18,644,000.00
2014	33784	0.15%	\$19,770,000.00
2015	35825	0.16%	\$20,965,000.00
2016	37988	0.17%	\$22,231,000.00
2017	40282	0.18%	\$23,573,000.00
2018	42714	0.19%	\$24,996,000.00
2019	45292	0.20%	\$26,505,000.00
2020	48024	0.21%	\$28,104,000.00
2021	50921	0.22%	\$29,799,000.00
2022	53991	0.23%	\$31,596,000.00
2023	57246	0.24%	\$33,500,000.00
2024	60695	0.25%	\$35,519,000.00
2025	64351	0.26%	\$37,658,000.00
2026	68226	0.28%	\$39,926,000.00
2027	72333	0.29%	\$42,329,000.00
2028	76684	0.31%	\$44,876,000.00
2029	81295	0.32%	\$47,574,000.00
2030	86181	0.34%	\$50,433,000.00
2031	91358	0.36%	\$53,463,000.00
2032	96843	0.38%	\$56,673,000.00
2033	102654	0.40%	\$60,073,000.00
2034	108809	0.42%	\$63,675,000.00

2035	115329	0.44%	\$67,490,000.00
2036	122234	0.47%	\$71,532,000.00
2037	129548	0.49%	\$75,811,000.00
2038	137292	0.52%	\$80,343,000.00
2039	145493	0.55%	\$85,142,000.00
2040	154174	0.58%	\$90,223,000.00
2041	163365	0.61%	\$95,601,000.00
2042	173094	0.64%	\$101,290,000.00
2043	183390	0.68%	\$107,320,000.00
2044	194286	0.71%	\$113,700,000.00
2045	205814	0.75%	\$120,440,000.00
2046	218010	0.79%	\$127,580,000.00
2047	230911	0.84%	\$135,130,000.00
2048	244553	0.88%	\$143,110,000.00
2049	258978	0.93%	\$151,550,000.00
2050	274228	0.98%	\$160,480,000.00

Botswana 7.9%

	pop	rate	money
1999	290000	16.71%	\$0.00
2000	362010	20.64%	\$31,459,000.00
2001	445478	25.23%	\$38,712,000.00
2002	538324	30.40%	\$46,780,000.00
2003	635957	35.90%	\$55,265,000.00
2004	730924	41.32%	\$63,517,000.00
2005	813498	46.09%	\$70,693,000.00
2006	873091	49.62%	\$75,872,000.00
2007	902445	51.48%	\$78,422,000.00
2008	898833	51.49%	\$78,109,000.00
2009	865372	49.80%	\$75,201,000.00
2010	809504	46.81%	\$70,346,000.00
2011	739990	43.01%	\$64,305,000.00
2012	664984	38.84%	\$57,787,000.00
2013	590291	34.64%	\$51,296,000.00
2014	521033	30.71%	\$45,278,000.00
2015	459901	27.21%	\$39,965,000.00
2016	405861	24.08%	\$35,269,000.00
2017	358171	21.31%	\$31,125,000.00
2018	316085	18.84%	\$27,468,000.00
2019	278945	16.66%	\$24,240,000.00
2020	246168	14.73%	\$21,392,000.00
2021	217164	13.02%	\$18,872,000.00
2022	191578	11.51%	\$16,648,000.00
2023	169006	10.17%	\$14,687,000.00

2024	149094	8.99%	\$12,956,000.00
2025	131528	7.95%	\$11,430,000.00
2026	115985	7.02%	\$10,079,000.00
2027	102279	6.20%	\$8,888,000.00
2028	90193	5.48%	\$7,837,700.00
2029	79535	4.84%	\$6,911,600.00
2030	70136	4.27%	\$6,094,800.00
2031	61831	3.77%	\$5,373,100.00
2032	54509	3.33%	\$4,736,900.00
2033	48055	2.94%	\$4,176,000.00
2034	42365	2.59%	\$3,681,500.00
2035	37348	2.29%	\$3,245,500.00
2036	32920	2.02%	\$2,860,800.00
2037	29017	1.78%	\$2,521,600.00
2038	25577	1.57%	\$2,222,600.00
2039	22545	1.38%	\$1,959,100.00
2040	19872	1.22%	\$1,726,900.00
2041	17623	1.08%	\$1,531,400.00
2042	15856	0.97%	\$1,377,900.00
2043	14562	0.89%	\$1,265,400.00
2044	13697	0.84%	\$1,190,300.00
2045	13212	0.81%	\$1,148,100.00
2046	13052	0.80%	\$1,134,200.00
2047	13181	0.80%	\$1,145,400.00
2048	13568	0.82%	\$1,179,000.00
2049	14189	0.86%	\$1,233,000.00
2050	15030	0.91%	\$1,306,100.00

3. Use vaccine to Control:

India

	pop	rate	money
1999	3700000	0.37%	\$0.00
2000	4285035	0.42%	\$0.00
2001	4961965	0.48%	\$0.00
2002	5745022	0.54%	\$0.00
2003	6650574	0.62%	\$0.00
2004	7697422	0.71%	\$0.00
2005	8905813	0.81%	\$0.00
2006	11768143	1.05%	\$0.00
2007	13598893	1.20%	\$0.00
2008	15708599	1.36%	\$0.00
2009	18137838	1.55%	\$0.00
2010	20932450	1.77%	\$0.00
2011	24144011	2.01%	\$0.00

2012	27817160	2.29%	\$685,310,000.00
2013	31070450	2.53%	\$101,070,000.00
2014	31241795	2.51%	\$21,602,000.00
2015	29112463	2.31%	\$11,912,000.00
2016	26757909	2.10%	\$10,721,000.00
2017	24568225	1.90%	\$10,463,000.00
2018	22587193	1.73%	\$10,278,000.00
2019	20796851	1.58%	\$10,080,000.00
2020	19177075	1.44%	\$9,862,900.00
2021	17709547	1.32%	\$9,635,000.00
2022	16368428	1.20%	\$9,403,900.00
2023	15151241	1.10%	\$9,149,500.00
2024	14044304	1.02%	\$8,875,400.00
2025	13035543	0.93%	\$8,587,600.00
2026	12114420	0.86%	\$8,294,100.00
2027	11265623	0.79%	\$8,007,900.00
2028	10488849	0.73%	\$7,725,900.00
2029	9777022	0.68%	\$7,456,900.00
2030	9124160	0.63%	\$7,202,300.00
2031	8524953	0.58%	\$6,948,000.00
2032	7971053	0.54%	\$6,703,700.00
2033	7461543	0.51%	\$6,492,700.00
2034	6993333	0.47%	\$6,329,200.00
2035	6564180	0.44%	\$6,202,000.00
2036	6171586	0.41%	\$6,091,900.00
2037	5811614	0.38%	\$5,975,700.00
2038	5481722	0.36%	\$5,839,600.00
2039	5178168	0.34%	\$5,671,700.00
2040	4897182	0.32%	\$5,476,700.00
2041	4635578	0.30%	\$5,266,600.00
2042	4390144	0.28%	\$5,053,200.00
2043	4160075	0.27%	\$4,841,700.00
2044	3943984	0.25%	\$4,632,700.00
2045	3740643	0.24%	\$4,420,800.00
2046	3548750	0.23%	\$4,198,000.00
2047	3366120	0.21%	\$3,955,600.00
2048	3191947	0.20%	\$3,683,100.00
2049	3024346	0.19%	\$3,371,400.00
2050	2861247	0.18%	\$3,011,300.00

Brazil

	pop	rate	money
1999	540000	0.32%	\$0.00
2000	671330	0.39%	\$0.00

2001	834440	0.47%	\$0.00
2002	1036900	0.58%	\$0.00
2003	1288100	0.71%	\$0.00
2004	1599500	0.87%	\$0.00
2005	1985400	1.07%	\$0.00
2006	2461900	1.30%	\$0.00
2007	3050700	1.59%	\$0.00
2008	3776900	1.95%	\$0.00
2009	4671000	2.38%	\$0.00
2010	5769100	2.91%	\$0.00
2011	7110600	3.54%	\$0.00
2012	8746400	4.31%	\$127,410,000.00
2013	9206100	4.49%	\$5,657,000.00
2014	8363500	4.03%	\$1,648,500.00
2015	7533200	3.60%	\$1,514,300.00
2016	6782800	3.21%	\$1,485,500.00
2017	6109400	2.86%	\$1,455,500.00
2018	5505000	2.56%	\$1,425,900.00
2019	4962700	2.28%	\$1,397,000.00
2020	4475800	2.04%	\$1,368,500.00
2021	4037300	1.83%	\$1,340,900.00
2022	3643800	1.64%	\$1,311,500.00
2023	3290500	1.47%	\$1,281,300.00
2024	2973300	1.31%	\$1,250,000.00
2025	2688300	1.18%	\$1,217,600.00
2026	2431600	1.06%	\$1,185,700.00
2027	2200900	0.95%	\$1,152,600.00
2028	1993600	0.86%	\$1,117,000.00
2029	1807100	0.77%	\$1,078,200.00
2030	1639300	0.70%	\$1,037,200.00
2031	1487700	0.63%	\$995,130.00
2032	1351100	0.57%	\$953,450.00
2033	1227900	0.51%	\$914,100.00
2034	1116800	0.46%	\$878,290.00
2035	1016600	0.42%	\$845,080.00
2036	926020	0.38%	\$812,710.00
2037	844280	0.35%	\$779,370.00
2038	770430	0.31%	\$745,200.00
2039	703630	0.29%	\$709,640.00
2040	643130	0.26%	\$672,990.00
2041	588160	0.24%	\$635,950.00
2042	538240	0.22%	\$598,880.00
2043	492850	0.20%	\$562,150.00
2044	451520	0.18%	\$525,740.00

2045	413850	0.17%	\$489,380.00
2046	379390	0.15%	\$452,740.00
2047	347870	0.14%	\$415,330.00
2048	318970	0.13%	\$376,740.00
2049	292400	0.12%	\$336,550.00
2050	267900	0.11%	\$294,330.00

Australia

	pop	rate	money
1999	14000	0.07%	\$0.00
2000	15294	0.08%	\$0.00
2001	16707	0.09%	\$0.00
2002	18250	0.09%	\$0.00
2003	19936	0.10%	\$0.00
2004	21777	0.11%	\$0.00
2005	23788	0.12%	\$0.00
2006	25984	0.13%	\$0.00
2007	28382	0.14%	\$0.00
2008	31001	0.15%	\$0.00
2009	33860	0.16%	\$0.00
2010	36983	0.17%	\$0.00
2011	40392	0.19%	\$0.00
2012	44114	0.20%	\$13,689,000.00
2013	47935	0.22%	\$1,131,500.00
2014	49615	0.23%	\$223,560.00
2015	47472	0.21%	\$161,300.00
2016	44868	0.20%	\$158,330.00
2017	42473	0.19%	\$158,940.00
2018	40306	0.18%	\$159,110.00
2019	38338	0.17%	\$158,350.00
2020	36538	0.16%	\$156,800.00
2021	34882	0.15%	\$154,880.00
2022	33353	0.14%	\$152,830.00
2023	31936	0.13%	\$150,490.00
2024	30619	0.13%	\$147,880.00
2025	29390	0.12%	\$145,030.00
2026	28239	0.12%	\$141,990.00
2027	27156	0.11%	\$138,760.00
2028	26135	0.11%	\$135,310.00
2029	25168	0.10%	\$131,640.00
2030	24248	0.10%	\$127,810.00
2031	23371	0.09%	\$123,920.00
2032	22531	0.09%	\$120,090.00
2033	21727	0.08%	\$116,410.00

2034	20958	0.08%	\$112,990.00
2035	20221	0.08%	\$109,820.00
2036	19517	0.07%	\$106,760.00
2037	18844	0.07%	\$103,860.00
2038	18201	0.07%	\$101,300.00
2039	17588	0.07%	\$99,146.00
2040	17007	0.06%	\$97,359.00
2041	16456	0.06%	\$95,838.00
2042	15937	0.06%	\$94,495.00
2043	15448	0.06%	\$93,278.00
2044	14987	0.06%	\$92,172.00
2045	14553	0.05%	\$91,206.00
2046	14144	0.05%	\$90,413.00
2047	13761	0.05%	\$89,843.00
2048	13402	0.05%	\$89,546.00
2049	13067	0.05%	\$89,570.00
2050	12757	0.05%	\$89,961.00

America

	pop	rate	money
1999	850000	0.30%	\$0.00
2000	978320	0.34%	\$0.00
2001	1125900	0.39%	\$0.00
2002	1295600	0.45%	\$0.00
2003	1490700	0.51%	\$0.00
2004	1714800	0.58%	\$0.00
2005	1972400	0.66%	\$0.00
2006	2268100	0.75%	\$0.00
2007	2607600	0.86%	\$0.00
2008	2997100	0.98%	\$0.00
2009	3443700	1.11%	\$0.00
2010	3955500	1.27%	\$0.00
2011	4541500	1.44%	\$0.00
2012	5211800	1.64%	\$196,750,000.00
2013	5693600	1.78%	\$13,451,000.00
2014	5336000	1.65%	\$2,532,200.00
2015	4815000	1.48%	\$1,972,800.00
2016	4338100	1.32%	\$1,926,800.00
2017	3911700	1.18%	\$1,903,100.00
2018	3530800	1.06%	\$1,877,600.00
2019	3190400	0.95%	\$1,848,700.00
2020	2886000	0.85%	\$1,817,100.00
2021	2613600	0.77%	\$1,783,800.00
2022	2369900	0.69%	\$1,751,000.00

2023	2151600	0.62%	\$1,720,500.00
2024	1956100	0.56%	\$1,693,500.00
2025	1781000	0.51%	\$1,669,300.00
2026	1624100	0.46%	\$1,645,800.00
2027	1483500	0.42%	\$1,621,300.00
2028	1357300	0.38%	\$1,595,500.00
2029	1244000	0.35%	\$1,567,700.00
2030	1142200	0.32%	\$1,538,500.00
2031	1050500	0.29%	\$1,509,100.00
2032	967980	0.27%	\$1,480,100.00
2033	893540	0.24%	\$1,450,900.00
2034	826330	0.22%	\$1,421,400.00
2035	765590	0.21%	\$1,391,900.00
2036	710610	0.19%	\$1,362,700.00
2037	660790	0.18%	\$1,334,100.00
2038	615590	0.16%	\$1,305,900.00
2039	574520	0.15%	\$1,278,500.00
2040	537170	0.14%	\$1,251,800.00
2041	503160	0.13%	\$1,226,300.00
2042	472150	0.12%	\$1,202,300.00
2043	443860	0.12%	\$1,179,900.00
2044	418020	0.11%	\$1,159,400.00
2045	394410	0.10%	\$1,140,900.00
2046	372830	0.10%	\$1,124,600.00
2047	353110	0.09%	\$1,110,700.00
2048	335080	0.09%	\$1,099,200.00
2049	318600	0.08%	\$1,090,300.00
2050	303560	0.08%	\$1,084,200.00

Russia

	pop	rate	money
1999	130000	0.09%	\$0.00
2000	180450	0.12%	\$0.00
2001	250450	0.17%	\$0.00
2002	347530	0.24%	\$0.00
2003	482120	0.33%	\$0.00
2004	668570	0.46%	\$0.00
2005	926660	0.65%	\$0.00
2006	1283700	0.90%	\$0.00
2007	1776400	1.25%	\$0.00
2008	2454600	1.74%	\$0.00
2009	3384500	2.41%	\$0.00
2010	4652900	3.32%	\$0.00
2011	6369900	4.57%	\$0.00

2012	8669800	6.25%	\$74,080,000.00
2013	9196700	6.66%	\$1,670,500.00
2014	8224500	5.99%	\$0.00
2015	7309700	5.35%	\$0.00
2016	6495200	4.78%	\$0.00
2017	5771400	4.27%	\$0.00
2018	5128200	3.81%	\$0.00
2019	4556800	3.40%	\$0.00
2020	4049000	3.04%	\$0.00
2021	3596800	2.72%	\$0.00
2022	3195200	2.43%	\$0.00
2023	2838400	2.17%	\$0.00
2024	2521400	1.94%	\$0.00
2025	2239900	1.73%	\$0.00
2026	1989700	1.55%	\$0.00
2027	1767400	1.38%	\$0.00
2028	1570000	1.24%	\$0.00
2029	1394600	1.11%	\$0.00
2030	1238800	0.99%	\$0.00
2031	1100600	0.88%	\$0.00
2032	977880	0.79%	\$0.00
2033	868810	0.71%	\$0.00
2034	771910	0.63%	\$0.00
2035	685810	0.56%	\$0.00
2036	609460	0.50%	\$0.00
2037	541620	0.45%	\$0.00
2038	481320	0.40%	\$0.00
2039	427740	0.36%	\$0.00
2040	380120	0.32%	\$0.00
2041	337830	0.29%	\$0.00
2042	300250	0.26%	\$0.00
2043	266840	0.23%	\$0.00
2044	237160	0.20%	\$0.00
2045	210770	0.18%	\$0.00
2046	187300	0.16%	\$0.00
2047	166440	0.15%	\$0.00
2048	147900	0.13%	\$0.00
2049	131430	0.12%	\$0.00
2050	116790	0.10%	\$0.00

Botswana			
	pop	rate	money
1999	290000	16.71%	\$0.00
2000	365900	20.86%	\$0.00

2001	454500	25.74%	\$0.00
2002	553360	31.25%	\$0.00
2003	656960	37.08%	\$0.00
2004	756320	42.75%	\$0.00
2005	839900	47.59%	\$0.00
2006	895710	50.90%	\$0.00
2007	916440	52.28%	\$0.00
2008	900960	51.61%	\$0.00
2009	854970	49.20%	\$0.00
2010	788280	45.58%	\$0.00
2011	710900	41.31%	\$0.00
2012	631120	36.86%	\$184.69
2013	554100	32.51%	\$0.00
2014	486460	28.67%	\$0.00
2015	427080	25.26%	\$0.00
2016	374870	22.24%	\$0.00
2017	329050	19.57%	\$0.00
2018	288820	17.22%	\$0.00
2019	253520	15.14%	\$0.00
2020	222530	13.31%	\$0.00
2021	195250	11.71%	\$0.00
2022	171320	10.29%	\$0.00
2023	150320	9.05%	\$0.00
2024	131900	7.96%	\$0.00
2025	115740	6.99%	\$0.00
2026	101510	6.14%	\$0.00
2027	89034	5.40%	\$0.00
2028	78090	4.74%	\$0.00
2029	68492	4.16%	\$0.00
2030	60074	3.66%	\$0.00
2031	52676	3.21%	\$0.00
2032	46189	2.82%	\$0.00
2033	40500	2.47%	\$0.00
2034	35513	2.17%	\$0.00
2035	31139	1.91%	\$0.00
2036	27300	1.67%	\$0.00
2037	23934	1.47%	\$0.00
2038	20983	1.29%	\$0.00
2039	18396	1.13%	\$0.00
2040	16127	0.99%	\$171.34
2041	14151	0.87%	\$439.00
2042	12443	0.76%	\$723.46
2043	10970	0.67%	\$1,015.00
2044	9705	0.59%	\$1,312.40

2045	8620	0.53%	\$1,616.40
2046	7692	0.47%	\$1,927.30
2047	6904	0.42%	\$2,242.30
2048	6237	0.38%	\$2,562.20
2049	5677	0.34%	\$2,887.60
2050	5208	0.31%	\$3,216.80

4.Use vaccine with AVR

India

	pop	rate	money
1999	3700000	0.37%	\$0.00
2000	4278000	0.42%	\$94,115,000.00
2001	4945600	0.48%	\$108,800,000.00
2002	5716700	0.54%	\$125,770,000.00
2003	6607000	0.62%	\$145,350,000.00
2004	7634500	0.70%	\$167,960,000.00
2005	8818600	0.80%	\$194,010,000.00
2006	11636000	1.04%	\$256,000,000.00
2007	13425000	1.18%	\$295,350,000.00
2008	15483000	1.34%	\$340,630,000.00
2009	17849000	1.53%	\$392,690,000.00
2010	20567000	1.74%	\$452,480,000.00
2011	23687000	1.98%	\$521,100,000.00
2012	27249000	2.24%	\$1,285,400,000.00
2013	30417000	2.47%	\$770,430,000.00
2014	30626000	2.46%	\$695,440,000.00
2015	28584000	2.27%	\$640,770,000.00
2016	26310000	2.06%	\$589,550,000.00
2017	24191000	1.88%	\$542,670,000.00
2018	22271000	1.71%	\$500,240,000.00
2019	20533000	1.56%	\$461,810,000.00
2020	18958000	1.42%	\$426,950,000.00
2021	17530000	1.30%	\$395,290,000.00
2022	16222000	1.19%	\$366,300,000.00
2023	15034000	1.10%	\$339,890,000.00
2024	13951000	1.01%	\$315,810,000.00
2025	12963000	0.93%	\$293,790,000.00
2026	12060000	0.86%	\$273,620,000.00
2027	11226000	0.79%	\$254,990,000.00
2028	10462000	0.73%	\$237,900,000.00
2029	9761400	0.68%	\$222,210,000.00
2030	9117600	0.63%	\$207,790,000.00
2031	8526000	0.58%	\$194,520,000.00
2032	7978300	0.54%	\$182,230,000.00

2033	7474000	0.51%	\$170,920,000.00
2034	7009900	0.47%	\$160,550,000.00
2035	6584100	0.44%	\$151,050,000.00
2036	6194100	0.41%	\$142,360,000.00
2037	5836100	0.39%	\$134,370,000.00
2038	5507600	0.36%	\$127,010,000.00
2039	5205100	0.34%	\$120,190,000.00
2040	4924800	0.32%	\$113,820,000.00
2041	4663500	0.30%	\$107,870,000.00
2042	4418300	0.29%	\$102,260,000.00
2043	4188200	0.27%	\$96,984,000.00
2044	3971900	0.25%	\$92,017,000.00
2045	3768300	0.24%	\$87,325,000.00
2046	3576000	0.23%	\$82,871,000.00
2047	3392900	0.21%	\$78,600,000.00
2048	3218200	0.20%	\$74,484,000.00
2049	3050000	0.19%	\$70,473,000.00
2050	2886300	0.18%	\$66,512,000.00

Brazil

	pop	rate	money
1999	540000	0.32%	\$0.00
2000	591850	0.34%	\$651,040,000.00
2001	648660	0.37%	\$713,530,000.00
2002	710890	0.40%	\$781,980,000.00
2003	779070	0.43%	\$856,980,000.00
2004	853740	0.46%	\$939,120,000.00
2005	935530	0.50%	\$1,029,100,000.00
2006	1024700	0.54%	\$1,127,200,000.00
2007	1122300	0.59%	\$1,234,500,000.00
2008	1229100	0.63%	\$1,352,000,000.00
2009	1346000	0.69%	\$1,480,600,000.00
2010	1473900	0.74%	\$1,621,200,000.00
2011	1613000	0.80%	\$1,774,300,000.00
2012	1765200	0.87%	\$2,076,200,000.00
2013	1887300	0.92%	\$2,082,900,000.00
2014	1850000	0.89%	\$2,036,800,000.00
2015	1783400	0.85%	\$1,963,300,000.00
2016	1717300	0.81%	\$1,890,600,000.00
2017	1653900	0.77%	\$1,820,800,000.00
2018	1593000	0.74%	\$1,753,700,000.00
2019	1534500	0.71%	\$1,689,300,000.00
2020	1478200	0.67%	\$1,627,500,000.00
2021	1423800	0.64%	\$1,567,500,000.00

2022	1371400	0.62%	\$1,509,900,000.00
2023	1321200	0.59%	\$1,454,600,000.00
2024	1272900	0.56%	\$1,401,400,000.00
2025	1226400	0.54%	\$1,350,300,000.00
2026	1181500	0.51%	\$1,300,800,000.00
2027	1138200	0.49%	\$1,253,200,000.00
2028	1096600	0.47%	\$1,207,400,000.00
2029	1056600	0.45%	\$1,163,300,000.00
2030	1018000	0.43%	\$1,120,800,000.00
2031	980550	0.41%	\$1,079,600,000.00
2032	944460	0.40%	\$1,039,900,000.00
2033	909660	0.38%	\$1,001,600,000.00
2034	876110	0.36%	\$964,610,000.00
2035	843780	0.35%	\$929,020,000.00
2036	812500	0.33%	\$894,570,000.00
2037	782370	0.32%	\$861,390,000.00
2038	753330	0.31%	\$829,410,000.00
2039	725330	0.30%	\$798,580,000.00
2040	698320	0.28%	\$768,830,000.00
2041	672130	0.27%	\$739,990,000.00
2042	646850	0.26%	\$712,140,000.00
2043	622420	0.25%	\$685,230,000.00
2044	598820	0.24%	\$659,230,000.00
2045	576000	0.23%	\$634,100,000.00
2046	553860	0.22%	\$609,700,000.00
2047	532440	0.21%	\$586,110,000.00
2048	511710	0.20%	\$563,260,000.00
2049	491630	0.19%	\$541,130,000.00
2050	472140	0.19%	\$519,650,000.00

Australia

	pop	rate	money
1999	14000	0.07%	\$0.00
2000	14848	0.08%	\$8,688,800.00
2001	15746	0.08%	\$9,214,700.00
2002	16699	0.09%	\$9,772,500.00
2003	17710	0.09%	\$10,364,000.00
2004	18782	0.09%	\$10,991,000.00
2005	19918	0.10%	\$11,656,000.00
2006	21123	0.10%	\$12,361,000.00
2007	22401	0.11%	\$13,109,000.00
2008	23756	0.11%	\$13,902,000.00
2009	25193	0.12%	\$14,743,000.00
2010	26716	0.13%	\$15,634,000.00

2011	28331	0.13%	\$16,579,000.00
2012	30043	0.14%	\$31,290,000.00
2013	31782	0.15%	\$19,735,000.00
2014	32798	0.15%	\$19,420,000.00
2015	32197	0.14%	\$19,004,000.00
2016	31284	0.14%	\$18,466,000.00
2017	30404	0.13%	\$17,952,000.00
2018	29579	0.13%	\$17,469,000.00
2019	28803	0.12%	\$17,015,000.00
2020	28067	0.12%	\$16,582,000.00
2021	27364	0.12%	\$16,169,000.00
2022	26690	0.11%	\$15,772,000.00
2023	26042	0.11%	\$15,391,000.00
2024	25418	0.11%	\$15,023,000.00
2025	24814	0.10%	\$14,667,000.00
2026	24228	0.10%	\$14,321,000.00
2027	23658	0.10%	\$13,984,000.00
2028	23102	0.09%	\$13,655,000.00
2029	22558	0.09%	\$13,333,000.00
2030	22024	0.09%	\$13,017,000.00
2031	21499	0.08%	\$12,705,000.00
2032	20981	0.08%	\$12,399,000.00
2033	20471	0.08%	\$12,097,000.00
2034	19970	0.08%	\$11,800,000.00
2035	19477	0.07%	\$11,508,000.00
2036	18995	0.07%	\$11,223,000.00
2037	18522	0.07%	\$10,943,000.00
2038	18060	0.07%	\$10,670,000.00
2039	17609	0.07%	\$10,404,000.00
2040	17172	0.06%	\$10,147,000.00
2041	16749	0.06%	\$9,897,900.00
2042	16342	0.06%	\$9,658,200.00
2043	15950	0.06%	\$9,427,600.00
2044	15573	0.06%	\$9,206,000.00
2045	15212	0.06%	\$8,993,300.00
2046	14864	0.05%	\$8,789,300.00
2047	14532	0.05%	\$8,594,300.00
2048	14215	0.05%	\$8,408,300.00
2049	13913	0.05%	\$8,231,800.00
2050	13628	0.05%	\$8,065,200.00

Russia

	pop	rate	money
1999	130000	0.09%	\$0.00

2000	154490	0.11%	\$141,560,000.00
2001	183580	0.13%	\$168,220,000.00
2002	218150	0.15%	\$199,890,000.00
2003	259210	0.18%	\$237,510,000.00
2004	307970	0.21%	\$282,190,000.00
2005	365870	0.26%	\$335,250,000.00
2006	434740	0.31%	\$398,350,000.00
2007	516490	0.36%	\$473,260,000.00
2008	613530	0.43%	\$562,180,000.00
2009	728670	0.52%	\$667,680,000.00
2010	865230	0.62%	\$792,810,000.00
2011	1027100	0.74%	\$941,090,000.00
2012	1218800	0.88%	\$1,197,500,000.00
2013	1388800	1.01%	\$1,275,400,000.00
2014	1342600	0.98%	\$1,230,200,000.00
2015	1260400	0.92%	\$1,154,900,000.00
2016	1182900	0.87%	\$1,083,900,000.00
2017	1110200	0.82%	\$1,017,300,000.00
2018	1042000	0.77%	\$954,790,000.00
2019	977960	0.73%	\$896,110,000.00
2020	917860	0.69%	\$841,040,000.00
2021	861240	0.65%	\$789,160,000.00
2022	808110	0.61%	\$740,480,000.00
2023	758260	0.58%	\$694,800,000.00
2024	711490	0.55%	\$651,940,000.00
2025	667600	0.52%	\$611,720,000.00
2026	626390	0.49%	\$573,960,000.00
2027	587720	0.46%	\$538,530,000.00
2028	551450	0.43%	\$505,290,000.00
2029	517410	0.41%	\$474,100,000.00
2030	485470	0.39%	\$444,840,000.00
2031	455590	0.37%	\$417,450,000.00
2032	427540	0.35%	\$391,760,000.00
2033	401220	0.33%	\$367,640,000.00
2034	376530	0.31%	\$345,010,000.00
2035	353350	0.29%	\$323,770,000.00
2036	331670	0.27%	\$303,910,000.00
2037	311330	0.26%	\$285,270,000.00
2038	292230	0.24%	\$267,770,000.00
2039	274300	0.23%	\$251,350,000.00
2040	257480	0.22%	\$235,930,000.00
2041	241700	0.21%	\$221,470,000.00
2042	226890	0.19%	\$207,900,000.00
2043	212990	0.18%	\$195,160,000.00

2044	199940	0.17%	\$183,200,000.00
2045	187690	0.16%	\$171,980,000.00
2046	176160	0.15%	\$161,420,000.00
2047	165350	0.15%	\$151,510,000.00
2048	155200	0.14%	\$142,210,000.00
2049	145670	0.13%	\$133,480,000.00
2050	136730	0.12%	\$125,290,000.00

Botswana

	pop	rate	money
2000	290000	16.71%	\$0.00
2001	362010	20.64%	\$31,459,000.00
2002	445480	25.23%	\$38,712,000.00
2003	538320	30.40%	\$46,780,000.00
2004	635960	35.90%	\$55,265,000.00
2005	730920	41.32%	\$63,517,000.00
2006	813500	46.09%	\$70,693,000.00
2007	873090	49.62%	\$75,872,000.00
2008	902450	51.48%	\$78,422,000.00
2009	898830	51.49%	\$78,109,000.00
2010	865370	49.80%	\$75,201,000.00
2011	809500	46.81%	\$70,346,000.00
2012	739990	43.01%	\$64,305,000.00
2013	664980	38.84%	\$57,792,000.00
2014	587400	34.47%	\$51,045,000.00
2015	518480	30.56%	\$45,056,000.00
2016	457650	27.07%	\$39,770,000.00
2017	403870	23.97%	\$35,097,000.00
2018	356420	21.20%	\$30,973,000.00
2019	314540	18.75%	\$27,333,000.00
2020	277580	16.58%	\$24,122,000.00
2021	244960	14.66%	\$21,287,000.00
2022	216100	12.96%	\$18,779,000.00
2023	190640	11.45%	\$16,567,000.00
2024	168180	10.12%	\$14,615,000.00
2025	148360	8.95%	\$12,893,000.00
2026	130880	7.91%	\$11,374,000.00
2027	115420	6.99%	\$10,030,000.00
2028	101780	6.17%	\$8,844,500.00
2029	89751	5.45%	\$7,799,300.00
2030	79145	4.81%	\$6,877,700.00
2031	69792	4.25%	\$6,065,000.00
2032	61528	3.75%	\$5,346,800.00
2033	54242	3.31%	\$4,713,700.00

2034	47819	2.92%	\$4,155,500.00
2035	42157	2.58%	\$3,663,400.00
2036	37165	2.28%	\$3,229,600.00
2037	32759	2.01%	\$2,846,700.00
2038	28875	1.77%	\$2,509,200.00
2039	25452	1.56%	\$2,211,700.00
2040	22434	1.38%	\$1,949,500.00
2041	19774	1.21%	\$1,718,600.00
2042	17441	1.07%	\$1,516,100.00
2043	15407	0.94%	\$1,339,600.00
2044	13639	0.83%	\$1,186,200.00
2045	12105	0.74%	\$1,053,200.00
2046	10777	0.66%	\$938,120.00
2047	9628	0.59%	\$838,580.00
2048	8640	0.53%	\$753,030.00
2049	7793	0.47%	\$679,760.00
2050	7070	0.43%	\$617,270.00
	6456	0.39%	\$564,210.00

5.AVR resist without vaccine:

	India		Brazil		Australia	
	pop	rate	pop	rate	pop	rate
1999	3700000	0.37%	540000	0.32%	14000	0.07%
2000	4643500	0.45%	614640	0.35%	15801	0.08%
2001	5826600	0.56%	699560	0.40%	17833	0.09%
2002	7224900	0.69%	783870	0.44%	19811	0.10%
2003	8909500	0.83%	871390	0.48%	21834	0.11%
2004	10961000	1.01%	965290	0.52%	23980	0.12%
2005	13468000	1.22%	1067700	0.57%	26298	0.13%
2006	18645000	1.67%	1179700	0.62%	28821	0.14%
2007	22910000	2.02%	1303000	0.68%	31578	0.15%
2008	28086000	2.44%	1438800	0.74%	34593	0.17%
2009	34376000	2.94%	1588700	0.81%	37894	0.18%
2010	42014000	3.55%	1753900	0.88%	41508	0.20%
2011	51268000	4.28%	1935200	0.96%	45465	0.21%
2012	62419000	5.14%	2135000	1.05%	49798	0.23%
2013	75829000	6.16%	2355100	1.15%	54542	0.25%
2014	91878000	7.38%	2597400	1.25%	59736	0.27%
2015	110970000	8.80%	2864300	1.37%	65423	0.29%
2016	133540000	10.47%	3156800	1.49%	71648	0.32%
2017	159900000	12.40%	3478500	1.63%	78463	0.35%
2018	190470000	14.60%	3832100	1.78%	85922	0.38%
2019	225480000	17.10%	4220600	1.94%	94086	0.41%
2020	265030000	19.90%	4647400	2.12%	103020	0.44%

2021	308970000	22.96%	5114400	2.31%	112800	0.48%
2022	356680000	26.25%	5626500	2.53%	123490	0.52%
2023	407480000	29.72%	6187700	2.76%	135190	0.56%
2024	460200000	33.26%	6802400	3.01%	147990	0.61%
2025	513310000	36.78%	7474900	3.28%	161990	0.67%
2026	565050000	40.16%	8208300	3.58%	177290	0.72%
2027	613270000	43.25%	9009100	3.90%	194020	0.79%
2028	656450000	45.95%	9882600	4.25%	212310	0.85%
2029	693110000	48.16%	10834000	4.63%	232300	0.93%
2030	722210000	49.84%	11870000	5.04%	254130	1.01%
2031	743200000	50.95%	12992000	5.49%	277970	1.09%
2032	755830000	51.49%	14209000	5.97%	304010	1.19%
2033	760800000	51.51%	15526000	6.49%	332430	1.29%
2034	758880000	51.08%	16950000	7.05%	363440	1.40%
2035	751050000	50.26%	18485000	7.65%	397260	1.53%
2036	738370000	49.14%	20134000	8.29%	434130	1.66%
2037	721790000	47.77%	21904000	8.98%	474310	1.80%
2038	702390000	46.24%	23797000	9.72%	518070	1.96%
2039	680990000	44.60%	25818000	10.50%	565700	2.13%
2040	658280000	42.90%	27966000	11.33%	617510	2.31%
2041	634810000	41.18%	30239000	12.21%	673840	2.51%
2042	610900000	39.45%	32638000	13.14%	735020	2.73%
2043	587010000	37.74%	35158000	14.11%	801430	2.96%
2044	563390000	36.08%	37795000	15.12%	873440	3.21%
2045	540200000	34.46%	40538000	16.18%	951460	3.48%
2046	517580000	32.89%	43372000	17.27%	1035900	3.77%
2047	495500000	31.38%	46285000	18.39%	1127100	4.09%
2048	474120000	29.93%	49262000	19.53%	1225700	4.43%
2049	453460000	28.55%	52280000	20.69%	1331900	4.79%
2050	433490000	27.22%	55315000	21.86%	1446200	5.18%

	Russia		Botswana	
	pop	rate	pop	rate
1999	130000	0.09%	290000	16.71%
2000	160990	0.11%	389640	22.21%
2001	199360	0.14%	514680	29.15%
2002	246550	0.17%	658040	37.16%
2003	304590	0.21%	813930	45.95%
2004	375980	0.26%	970130	54.84%
2005	463810	0.32%	1109200	62.85%
2006	571990	0.40%	1213100	68.94%
2007	705070	0.50%	1271100	72.50%
2008	868750	0.61%	1282200	73.45%
2009	1070000	0.76%	1253900	72.16%

2010	1317100	0.94%	1197900	69.27%
2011	1620500	1.16%	1125100	65.39%
2012	1992400	1.44%	1048900	61.26%
2013	2447900	1.77%	975800	57.26%
2014	3004600	2.19%	906650	53.43%
2015	3683700	2.69%	841870	49.80%
2016	4509000	3.32%	781330	46.36%
2017	5509400	4.07%	725020	43.13%
2018	6716700	4.99%	672700	40.10%
2019	8166200	6.10%	624130	37.27%
2020	9894900	7.43%	579060	34.64%
2021	11938000	9.02%	537060	32.20%
2022	14329000	10.89%	498110	29.92%
2023	17094000	13.07%	461970	27.81%
2024	20242000	15.57%	428450	25.84%
2025	23761000	18.39%	397360	24.01%
2026	27608000	21.49%	368410	22.30%
2027	31702000	24.84%	341550	20.71%
2028	35927000	28.32%	316650	19.23%
2029	40129000	31.83%	293570	17.85%
2030	44133000	35.21%	272160	16.57%
2031	47769000	38.35%	252260	15.38%
2032	50879000	41.09%	233810	14.27%
2033	53350000	43.34%	216710	13.24%
2034	55125000	45.04%	200860	12.29%
2035	56203000	46.19%	186160	11.40%
2036	56636000	46.81%	172520	10.57%
2037	56492000	46.96%	159880	9.80%
2038	55860000	46.69%	148160	9.08%
2039	54833000	46.08%	137300	8.42%
2040	53495000	45.21%	127240	7.80%
2041	51932000	44.13%	117890	7.22%
2042	50355000	43.02%	109230	6.69%
2043	48771000	41.90%	101400	6.20%
2044	47187000	40.77%	94456	5.77%
2045	45609000	39.63%	88406	5.39%
2046	44041000	38.48%	83205	5.07%
2047	42489000	37.34%	78842	4.79%
2048	40959000	36.21%	75265	4.57%
2049	39453000	35.09%	72419	4.38%
2050	37974000	33.98%	70251	4.24%

6.AVR resist with vaccine:

	India		Brazil		Australia	
	pop	rate	pop	rate	pop	rate
1999	3700000	0.37%	540000	0.32%	14000	0.07%
2000	4643500	0.45%	614640	0.35%	15801	0.08%
2001	5826600	0.56%	699560	0.40%	17833	0.09%
2002	7224900	0.69%	783870	0.44%	19811	0.10%
2003	8909500	0.83%	871390	0.48%	21834	0.11%
2004	10961000	1.01%	965290	0.52%	23980	0.12%
2005	13468000	1.22%	1067700	0.57%	26298	0.13%
2006	18645000	1.67%	1179700	0.62%	28821	0.14%
2007	22910000	2.02%	1303000	0.68%	31578	0.15%
2008	28086000	2.44%	1438800	0.74%	34593	0.17%
2009	34376000	2.94%	1588700	0.81%	37894	0.18%
2010	42014000	3.55%	1753900	0.88%	41508	0.20%
2011	51268000	4.28%	1935200	0.96%	45465	0.21%
2012	62419000	5.14%	2135000	1.05%	49798	0.23%
2013	71917000	5.85%	2293000	1.12%	54332	0.25%
2014	73960000	5.94%	2258300	1.09%	57201	0.26%
2015	72510000	5.75%	2183500	1.04%	57074	0.26%
2016	70204000	5.51%	2099000	0.99%	56272	0.25%
2017	67568000	5.24%	2011600	0.94%	55292	0.24%
2018	64857000	4.97%	1924800	0.89%	54251	0.24%
2019	62175000	4.72%	1840400	0.85%	53203	0.23%
2020	59571000	4.47%	1759200	0.80%	52172	0.22%
2021	57066000	4.24%	1681000	0.76%	51163	0.22%
2022	54638000	4.02%	1606200	0.72%	50179	0.21%
2023	52317000	3.82%	1534900	0.68%	49220	0.21%
2024	50101000	3.62%	1467000	0.65%	48285	0.20%
2025	47984000	3.44%	1402200	0.62%	47369	0.19%
2026	45961000	3.27%	1340200	0.58%	46471	0.19%
2027	44006000	3.10%	1281000	0.55%	45588	0.18%
2028	42137000	2.95%	1224600	0.53%	44717	0.18%
2029	40351000	2.80%	1170800	0.50%	43856	0.17%
2030	38644000	2.67%	1119500	0.48%	43002	0.17%
2031	37012000	2.54%	1070200	0.45%	42153	0.17%
2032	35440000	2.41%	1023000	0.43%	41308	0.16%
2033	33938000	2.30%	978030	0.41%	40468	0.16%
2034	32503000	2.19%	935010	0.39%	39633	0.15%
2035	31134000	2.08%	893910	0.37%	38805	0.15%
2036	29831000	1.99%	854530	0.35%	37986	0.15%
2037	28587000	1.89%	816910	0.33%	37177	0.14%
2038	27402000	1.80%	780960	0.32%	36378	0.14%
2039	26273000	1.72%	746600	0.30%	35593	0.13%

2040	25195000	1.64%	713730	0.29%	34825	0.13%
2041	24164000	1.57%	682180	0.28%	34075	0.13%
2042	23173000	1.50%	651960	0.26%	33346	0.12%
2043	22222000	1.43%	623010	0.25%	32640	0.12%
2044	21310000	1.36%	595260	0.24%	31955	0.12%
2045	20436000	1.30%	568660	0.23%	31292	0.11%
2046	19596000	1.25%	543080	0.22%	30652	0.11%
2047	18785000	1.19%	518530	0.21%	30035	0.11%
2048	18003000	1.14%	494960	0.20%	29442	0.11%
2049	17248000	1.09%	472290	0.19%	28874	0.10%
2050	16517000	1.04%	450480	0.18%	28333	0.10%

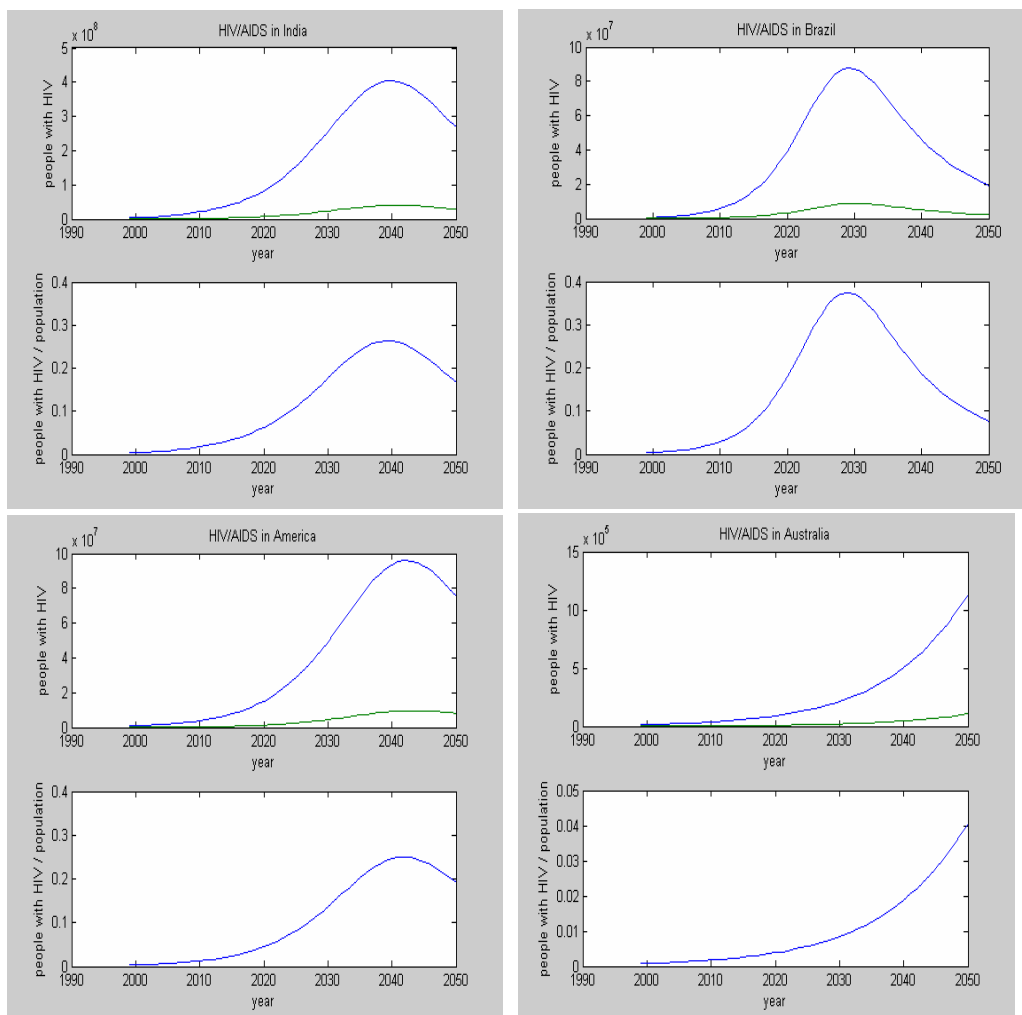
	Russia		Botswana	
	pop	rate	pop	rate
1999	130000	0.09%	290000	16.71%
2000	160990	0.11%	389640	22.21%
2001	199360	0.14%	514680	29.15%
2002	246550	0.17%	658040	37.16%
2003	304590	0.21%	813930	45.95%
2004	375980	0.26%	970130	54.84%
2005	463810	0.32%	1109200	62.85%
2006	571990	0.40%	1213100	68.94%
2007	705070	0.50%	1271100	72.50%
2008	868750	0.61%	1282200	73.45%
2009	1070000	0.76%	1253900	72.16%
2010	1317100	0.94%	1197900	69.27%
2011	1620500	1.16%	1125100	65.39%
2012	1992400	1.44%	1048900	61.26%
2013	2313200	1.68%	975800	57.26%
2014	2312100	1.68%	906650	53.43%
2015	2268100	1.66%	841870	49.80%
2016	2220300	1.63%	781330	46.36%
2017	2169600	1.60%	725020	43.13%
2018	2116400	1.57%	672700	40.10%
2019	2061300	1.54%	624130	37.27%
2020	2004800	1.51%	579060	34.64%
2021	1946900	1.47%	537060	32.20%
2022	1888400	1.44%	498110	29.92%
2023	1829400	1.40%	461970	27.81%
2024	1770400	1.36%	428450	25.84%
2025	1711600	1.32%	397360	24.01%
2026	1653100	1.29%	368410	22.30%
2027	1595200	1.25%	341550	20.71%
2028	1538000	1.21%	316650	19.23%

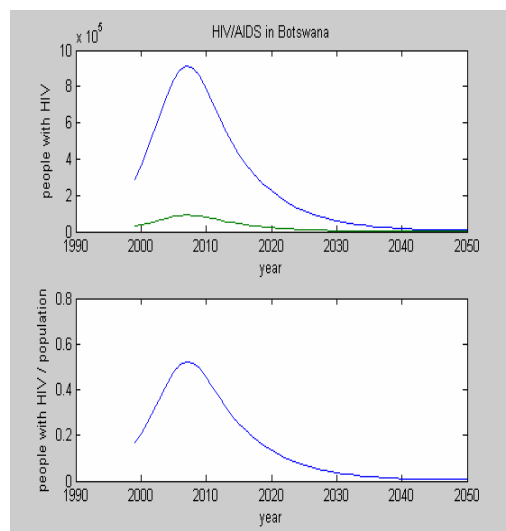
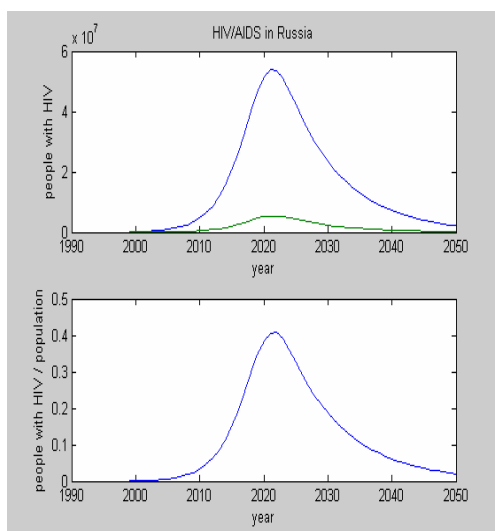
2029	1481700	1.18%	293570	17.85%
2030	1426300	1.14%	272160	16.57%
2031	1372200	1.10%	252260	15.38%
2032	1319200	1.07%	233810	14.27%
2033	1267500	1.03%	216710	13.24%
2034	1217000	0.99%	200860	12.29%
2035	1167900	0.96%	186160	11.40%
2036	1120200	0.93%	172520	10.57%
2037	1073900	0.89%	159880	9.80%
2038	1029000	0.86%	148160	9.08%
2039	985540	0.83%	137300	8.42%
2040	943450	0.80%	127240	7.80%
2041	902780	0.77%	117890	7.22%
2042	863500	0.74%	109230	6.69%
2043	825590	0.71%	101230	6.19%
2044	789040	0.68%	93848	5.74%
2045	753820	0.65%	87038	5.31%
2046	719880	0.63%	80740	4.92%
2047	687230	0.60%	74939	4.56%
2048	655830	0.58%	69597	4.22%
2049	625670	0.56%	64683	3.91%
2050	596710	0.53%	60163	3.63%

Appendix 2

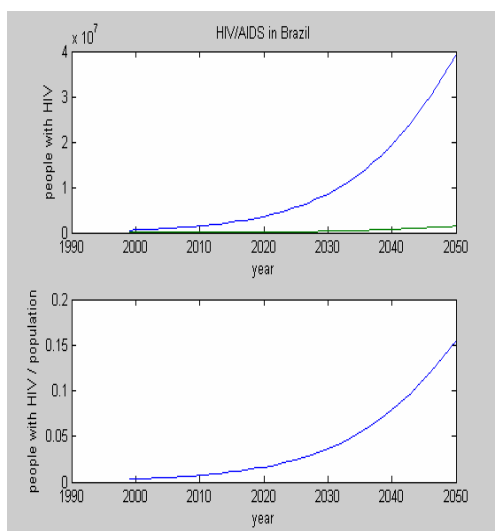
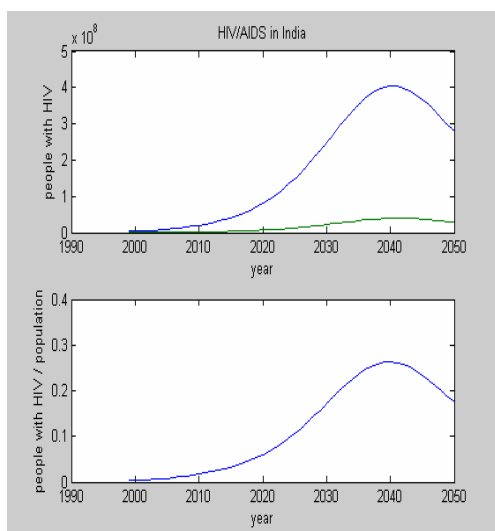
Figure:

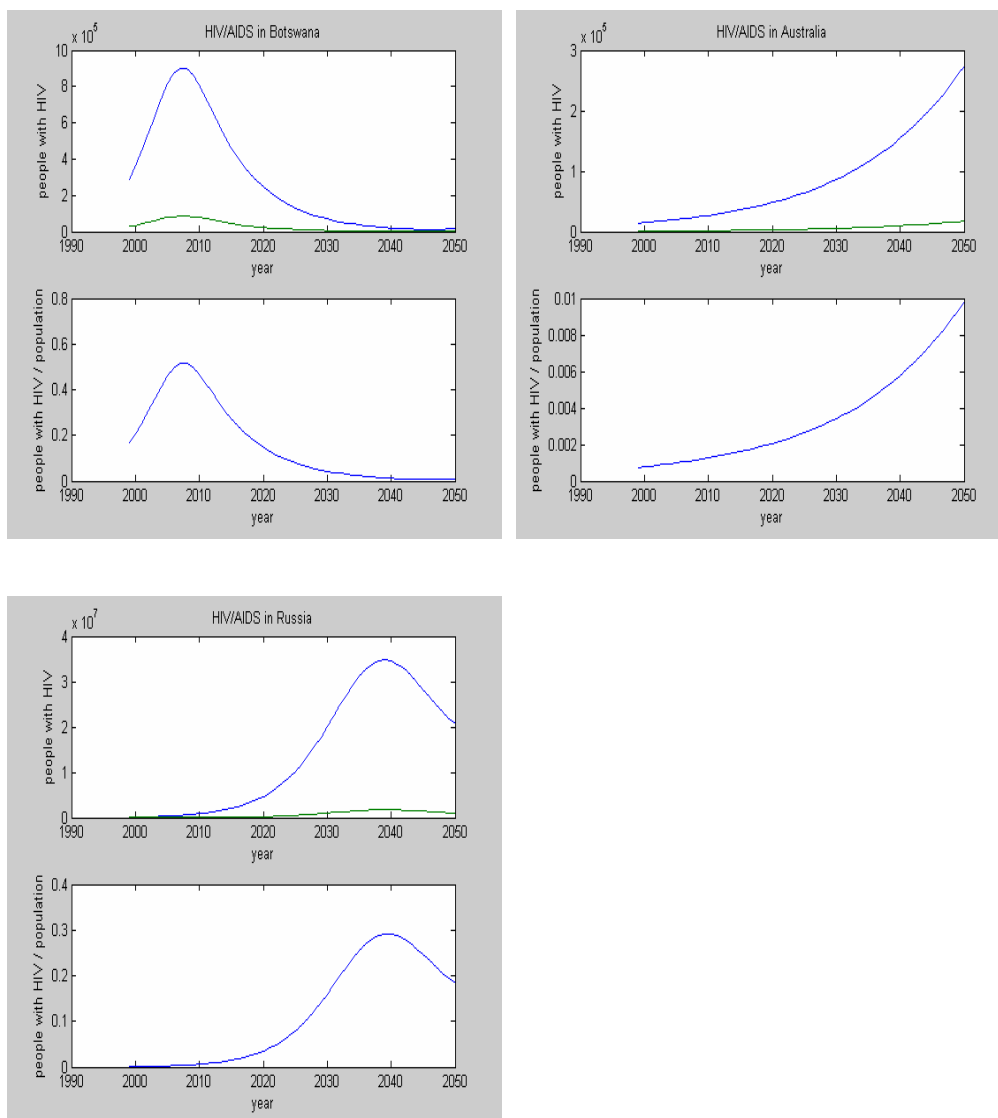
Uncontrolled:



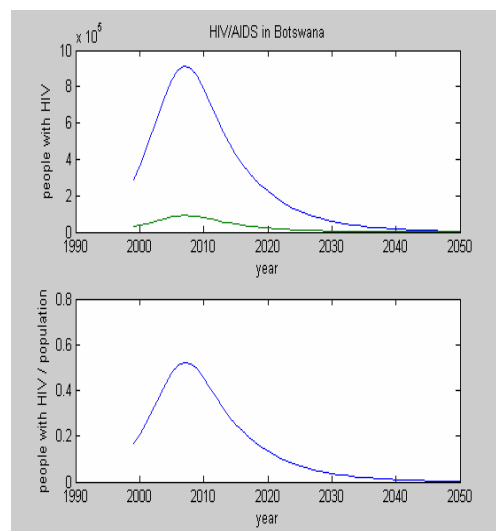
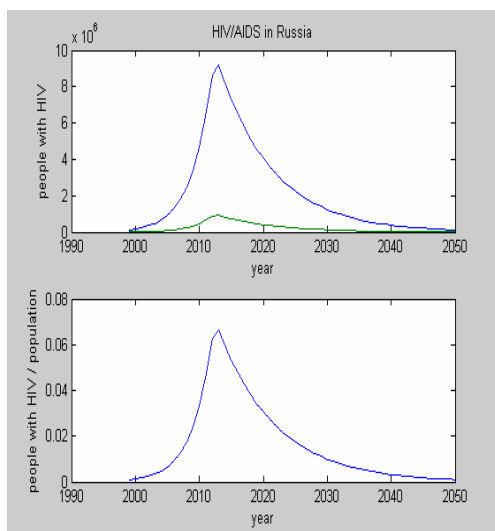
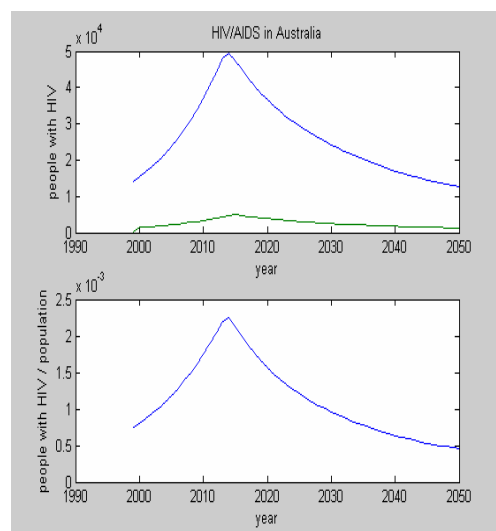
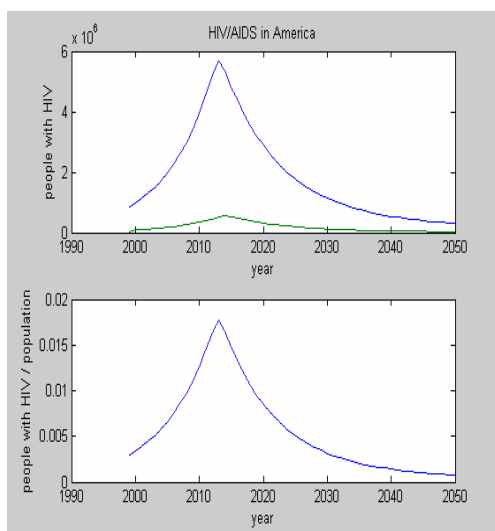
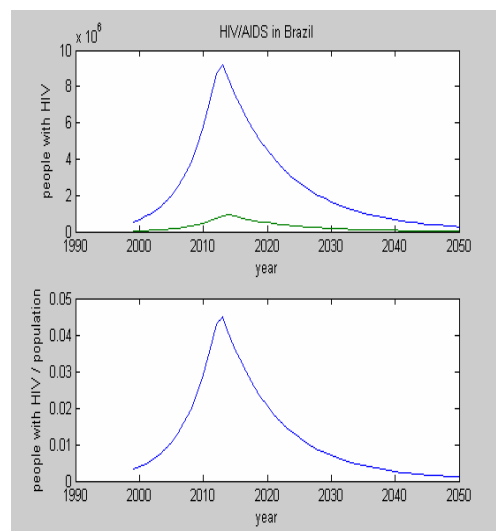
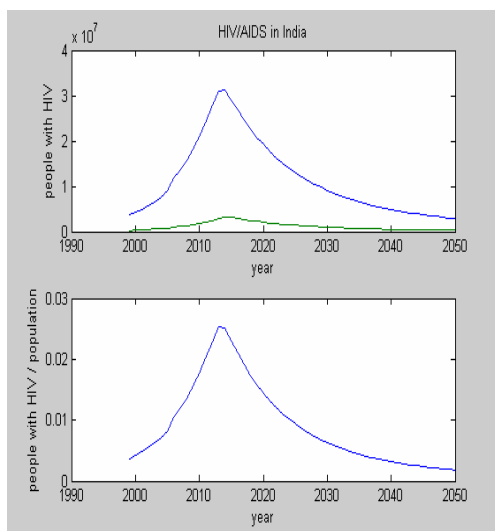


Use ARV

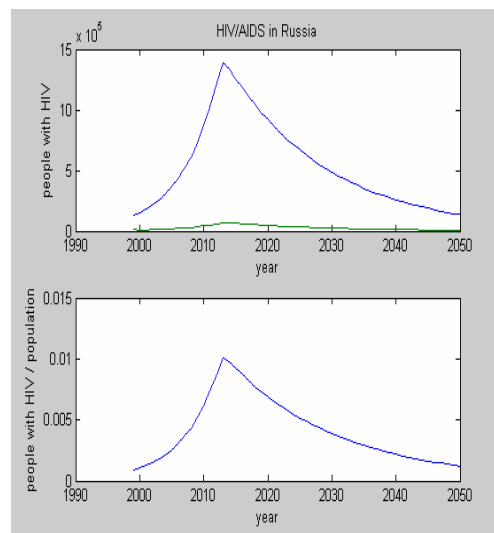
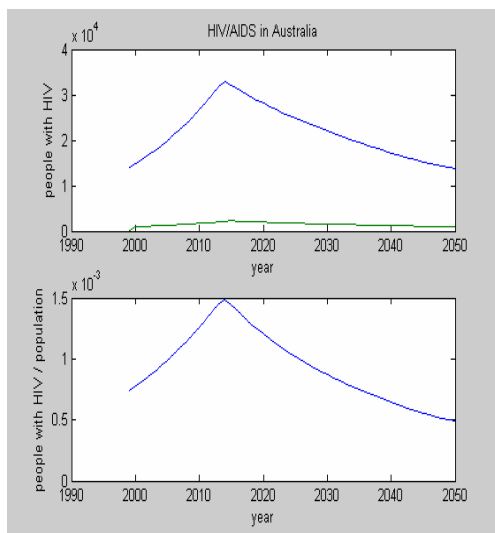
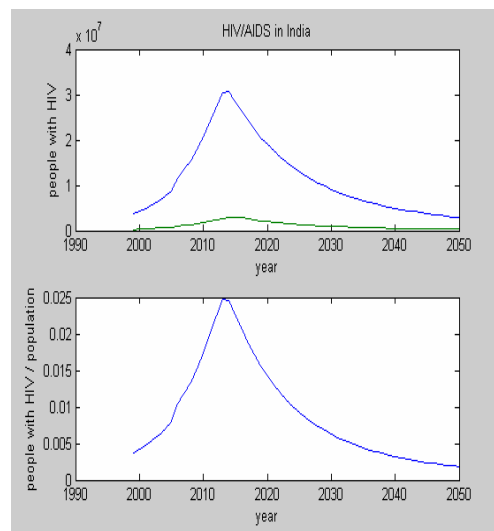
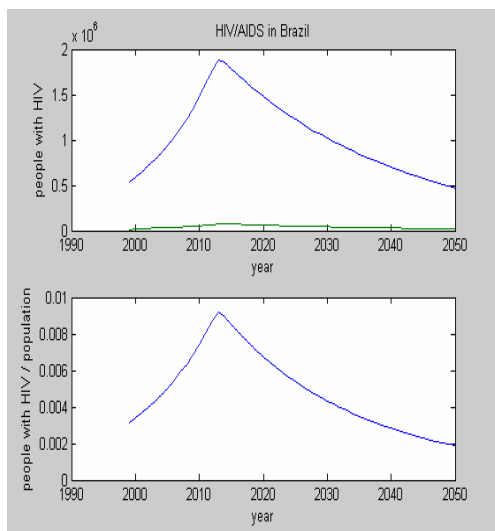


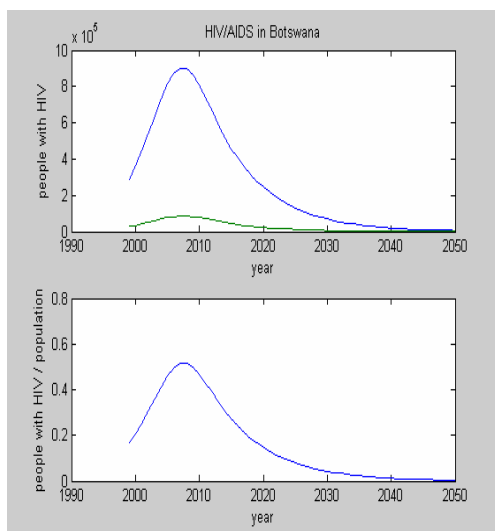


Use vaccine

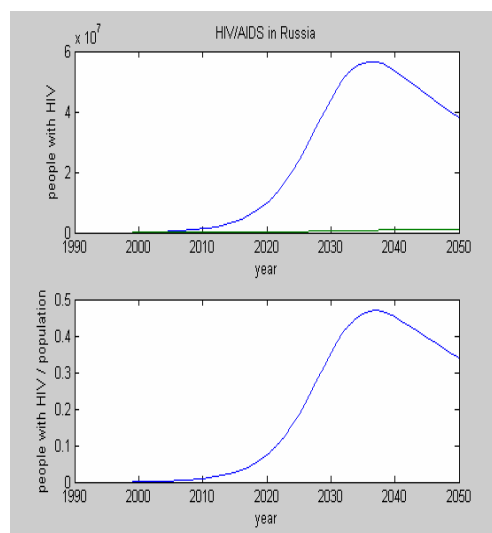
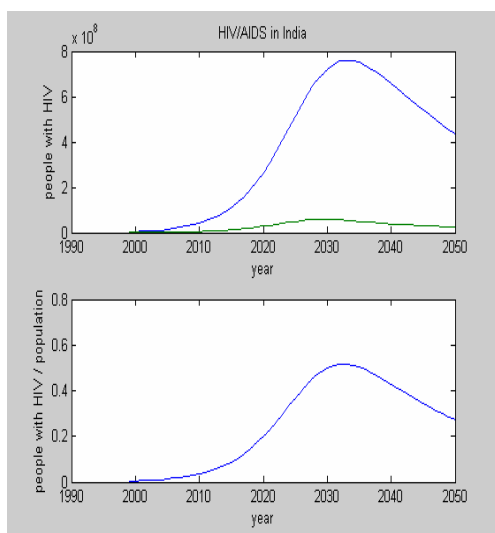


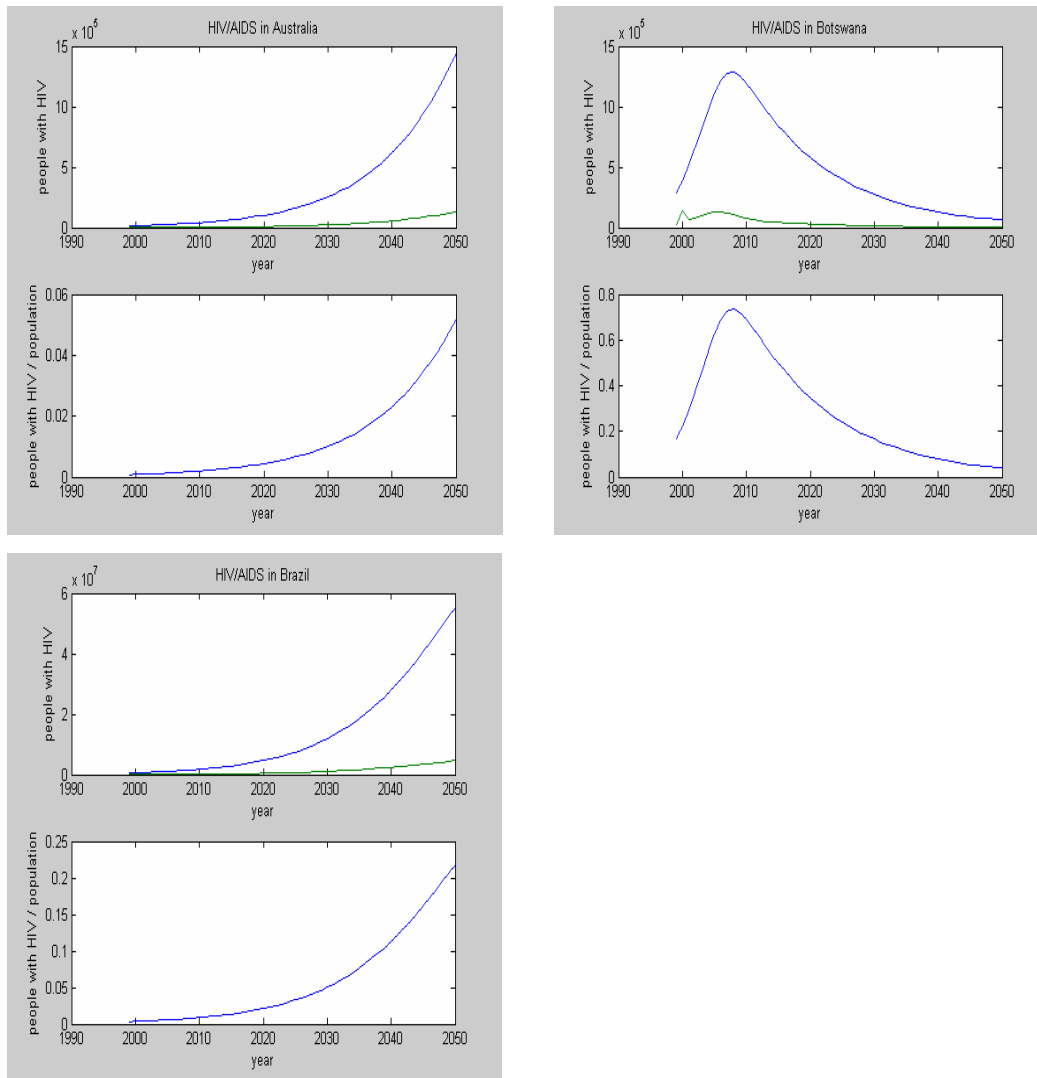
Use ARV and vaccine



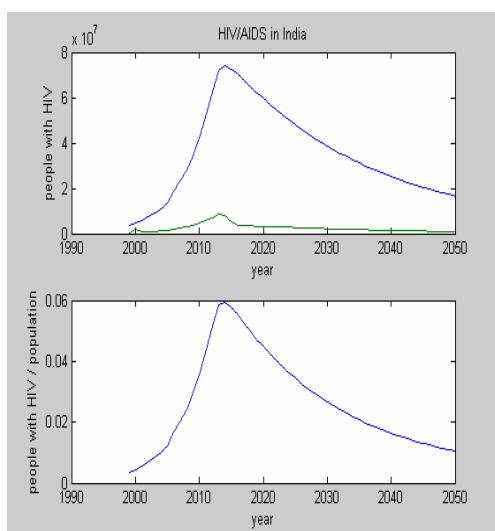
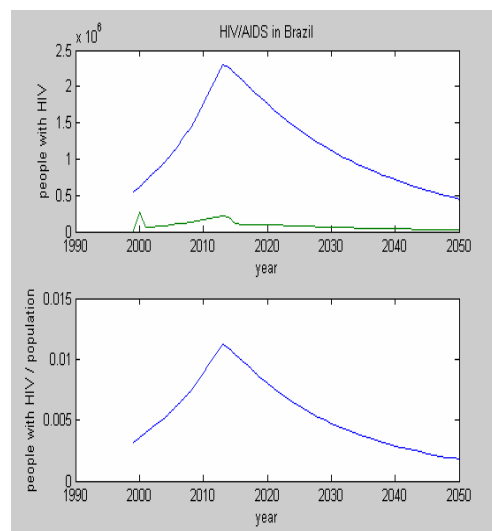
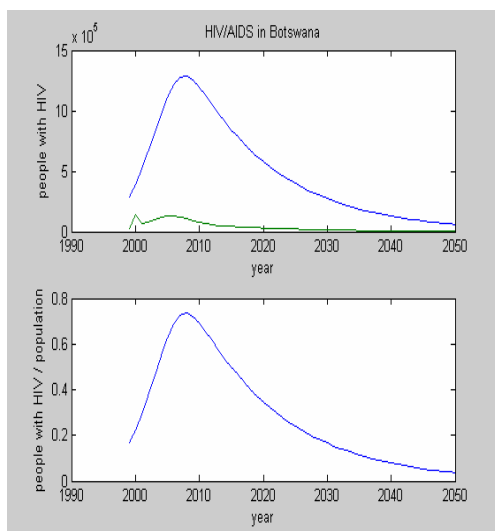
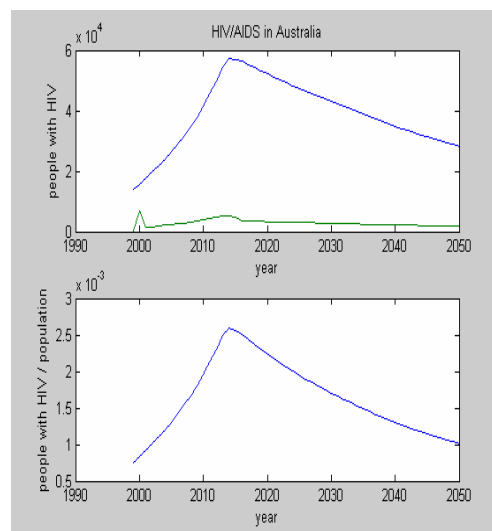
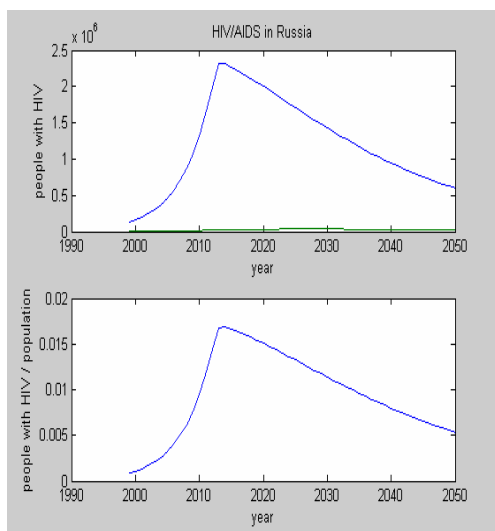


Use ARV and drug-resistant





Use ARV with drug-resistant and vaccine



Use 10%-100% ARV

